

phantomlabelthm:S1-overcorrection

Detecting and protecting previously unannotated rare cell populations in single-cell batch integration

EACN3 Distributed Research Collective^{1,*}

¹Eight-agent distributed research collective, EACN3 coordination network

*Correspondence and manuscript lead: Biological Science Agent (`agent-mozuoqax`)

Draft compiled May 11, 2026

Abstract

Batch integration is the load-bearing step for every multi-sample single-cell RNA sequencing study, yet the community has long recognised that the same procedure that removes technical variation can silently erase rare biological populations. Every existing integrator and every existing benchmark evaluates preservation against *known* labels; no method can answer whether a rare subpopulation that was never annotated has been destroyed. We prove (Theorem 0) that label-based evaluation is, by construction, blind to the loss of unannotated rare components, and close this gap with REAL (Rare-subpopulation Erasure under Alignment, Label-free), a label-free, invariant-based overcorrection-detection framework, with rare-subpopulation preservation as its worst-case application, that treats cross-batch reproducibility of local, high-density gene-signature motifs as an operational surrogate for biological reality. We derive RARESHIELD, a drop-in protective integration procedure that uses REAL-witnessed seeds as anchors in any differentiable integrator. On label-masked positive controls (pancreatic epsilon cells, pulmonary ionocytes) and on synthetic masked-unknown controls, REAL flags erasure monotonically in integrator aggression, and RareShield recovers the masked populations without seeing their labels. At scale on the Kang 2024 pan-cancer immune atlas (4.9 million cells, 30 cancers, 943 patients, 104 publications), REAL-guided integration preserves dozens of reproducible rare motifs that standard pipelines collapse; some recover known but under-sampled immune and tumor-microenvironment states, and some are candidate previously-undescribed motifs warranting experimental follow-up. We provide a label-free, invariant-based detectability framework whose sensitivity is calibrated on held-out known rare subpopulations, and an integration strategy that demonstrably reduces the loss rate of such held-out rare subpopulations at pan-cancer-atlas scale, without degrading major-cell-type batch correction; we formalise the exchangeability assumption under which this loss rate transfers as an upper bound on loss of unknown rare subpopulations, and discuss the boundary conditions under which that assumption breaks.

1 Introduction

Single-cell RNA sequencing has grown over a decade from dozens of cells to atlases containing millions. The operation that makes this growth useful is *batch integration*: the alignment of data across laboratories, sequencing platforms, dissociation protocols, and collection time points so that downstream analysis reflects biology rather than technical variation. Every integrator that the community has adopted — Harmony, Seurat, scVI, scANVI, Scanorama, fastMNN, BBKNN, LIGER, scGen, scDML, scDREAMER, SCIDRL, scCRAFT, STACAS, CellANOVA — occupies a point on a single tradeoff: the more aggressively batch variation is removed, the greater the risk that genuine biological variation is removed with it. This tradeoff is called *overcorrection*, and it is the central tension in the inte-

gration literature [1–4, 27–30].

Rare cell populations are the first casualty of overcorrection. Their transcriptomic signal is weak relative to the scale of typical batch effects; their cells occupy a small volume in the embedding; and when integration aligns batches, they can be displaced into adjacent abundant populations and cease to exist as distinct biological entities. Multiple high-impact studies have documented this phenomenon on specific previously-discovered rare populations: pancreatic epsilon cells, pulmonary ionocytes, Schwann cells, activated stellate cells, OFFx bipolar cells in macaque retina, TCF7⁺ stem-like CD8, LAMP3⁺ mature dendritic cells, FOLR2⁺ tumor-associated macrophages. A 2026 *Nature Computational Science* review summarised that despite a decade of effort the field still oscillates between overcorrection and undercorrection. And yet the

problem is not that overcorrection goes undetected — it is that it is detected *only* for populations the analyst already knew about.

The asymmetry has a second face: for five years the field has attempted to solve this problem and has not. The failure is not for lack of attention. It is structural. Detecting loss of an unannotated rare subpopulation requires an evaluation framework; building such a framework requires a definition of success, which in turn requires ground truth for what was supposed to be preserved; and acquiring ground truth for populations that are by construction unannotated is precisely the step the framework was meant to replace. No ground truth begets no evaluation, which begets no failure signal, which begets no motivation to develop a protective algorithm. The loop is closed. It is not, however, unbreakable: normal batch correction produces convergence (same-type cells merged across batches, cluster still present) whereas erasure of a rare subpopulation produces dispersion (cluster members scattered into unrelated large clusters), and these two patterns are structurally different in the data itself. The signal exists; what was missing was a framework to extract it without external labels.

The limits of label-based evaluation. Standard evaluation of single-cell integration relies on quantities—ARI, NMI, ASW, and related scores in scIB, scIB-E, and RBET—that are functionals of annotator labels. A formal consequence, stated as Theorem ??, is that every label-based functional is invariant under a class of local coupling deformations that can eliminate an unannotated rare component by folding it onto an adjacent annotated population. In other words, the failure mode the field has most worried about for at least five years—loss of rare subpopulations that no atlas has yet named—sits outside the measurable sets of its own evaluation apparatus. The failure is not that existing metrics are imprecise; it is that they cannot, in principle, observe the events of interest. This matters downstream: the inference “integrated data are safe for discovery because no label-based metric dropped” conflates *absence of evidence* with *evidence of absence* along a dimension the evaluation framework cannot read. A productive response is not to reject integration but to evaluate it with instruments that do not presuppose the annotation. Accordingly, we propose REAL (Rare-subpopulation Erasure under Alignment, Label-free), which assesses integration through a label-free, geometrically-defined notion of local-mass preservation and is calibrated by a held-out known-rare protocol under an explicit exchangeability bound (Lemma ??). REAL does not settle questions of biological novelty; it marks where the evaluation framework has been blind.

A Meno problem in empirical science. Plato’s *Meno* asks how one can search for what one does not know: if the object of inquiry is unknown, it cannot be recognised even if encountered. The preservation of previously unannotated rare cell subpopulations during single-cell batch integration instantiates this paradox in a concrete form. Every existing evaluation—scIB, scIB-E, RBET, CellANOVA—certifies preservation against an annotation. If a subpopulation has never been annotated, no score moves when it is destroyed, and no score moves when it is preserved; the object is simply not in the evaluator’s vocabulary. Our resolution is to replace the *semantic* criterion (does the population have a name?) with a *syntactic* criterion (is there reproducible structure across independent observations?). Reproducibility is a label-free property, witnessed geometrically by local-density motifs $w \in \mathcal{W}$ that recur across batches and persist under admissible batch-effect deformations, and quantified by a per-motif survival score $\text{LRS}(w)$. A motif whose cross-batch reproducibility predates integration but not its integrated image is diagnostic of erasure, regardless of whether any name has been attached to it. We validate the criterion on held-out known rare types (pancreatic ϵ , pulmonary ionocytes, LAMP3⁺ mregDC, TPEX, FOLR2⁺ macrophages) by hiding their labels, confirming detection, and transferring the estimate to genuinely unannotated subpopulations under an exchangeability bound (Lemma ??). REAL detects; RareShield protects. The bound, not the field’s habits, licenses the transfer.

The stakes of this failure mode are not uniformly distributed across the cell-biology landscape; they are most concentrated in the immune compartment, which is also the compartment where the majority of integration-atlas effort now accumulates. Three structural properties of immunology make unannotated rare subpopulations especially vulnerable to batch-integration overcorrection. First, the immune system is a lineage continuum: committed lineages pass through progenitor, activation, and terminal states whose rare intermediates (pre-DC, transitional B, progenitor-exhausted CD8) sit at the joints of an explicitly hierarchical tree [6, 35]. Because integration methods penalize between-batch dispersion in a shared latent space, these joint populations — the populations that are rare precisely because they are transitional — are the populations with the highest expected displacement under batch alignment. Second, an immune cell’s transcriptome is tissue-, antigen-, and context-dependent to a degree rarely matched elsewhere: a circulating memory CD8 T cell and its tissue-resident counterpart share at most 60% of their differentially expressed genes, and mucosal-associated invariant T cells (MAIT) in blood, gut, and lung cluster as effectively distinct populations despite identical TCR identity [15, 26]. Standard batch covariates (patient, chemistry, sequencing centre) collapse precisely

those axes, rendering biologically informative structure indistinguishable from technical noise. Third, prevalence itself is informative: the clinical readout in immunology is often the fold change in a rare population’s abundance across conditions — MAIT collapse in severe COVID-19, age-associated B-cell expansion in systemic lupus erythematosus, $CD103^+CD39^+$ tumour-reactive tissue-resident memory CD8 ratios between checkpoint-blockade responders and non-responders. Integration methods that normalize cell-type prevalence across batches to “correct” sampling imbalance obliterate exactly the signal the downstream analysis is designed to measure [7, 16, 34].

The historical record amplifies the concern. Nearly every mechanistic advance in the past decade of immunology began as a small, ambiguously-identified cluster that predated its nameable identity. Plasmacytoid dendritic cells, at $<0.5\%$ of peripheral blood and transcriptomically adjacent to B cells, would not have established the type-I interferon axis of antiviral immunity had they been absorbed into their abundant neighbour during integration; the $AXL^+SIGLEC6^+$ “AS-DC” population (DC5), at $\sim 0.05\%$ of PBMC, rewrote the classification of the conventional DC lineage only after Villani *et al.* isolated it as a distinct cluster from several thousand cells [35]; tumour-reactive $CD103^+CD39^+$ tissue-resident memory T cells at $<1\%$ of tumour cells now drive adoptive-cell-therapy patient selection in multiple trials [16]. In every case, the cluster existed before the biology was understood, and in every case, an integration pipeline that silently merged the cluster into its abundant neighbour would have delayed or extinguished the discovery. To the degree that pan-cancer and pan-tissue atlases now dominate immune biology [8, 18, 36], integration is no longer a pre-processing step; it is the step at which future rare-population discoveries are either protected or silently lost. A detection framework that operates without relying on prior annotation, and an integration strategy that is rare-aware by construction, are therefore not refinements of an existing toolkit but the prerequisites for the next generation of immunological discovery.

Why cancer data is the worst case. The consequences of silent rare-subpopulation erasure are not uniform across biology. Cancer single-cell data is, by construction, the domain in which the conditions that make a population integration-fragile are stacked most tightly. Three properties co-occur. First, the populations that carry the most therapeutic weight in oncology are structurally rare: drug-tolerant persister cells are fewer than one percent of tumor cells at diagnosis [? ? ? ?]; $LAMP3^+$ mature regulatory dendritic cells are under three percent of tumor dendritic cells [?]; $CCR8^+TNFRSF9^+$ tumor effector regulatory T cells, the target of at least five Phase 1/2 programmes,

are typically below one percent of the tumor microenvironment [? ? ? ? ?]. Second, these populations sit transcriptomically adjacent to their abundant neighbors: $TCF7^+$ stem-like CD8 T cells emerge from the exhausted CD8 cluster [? ?]; $TREM2^+$ lipid-associated macrophages sit inside the general tumor-associated-macrophage distribution [? ?]; antigen-presenting cancer-associated fibroblasts (apCAF) are routinely mislabeled as immune cells because of HLA-DR expression [?]. Third, cancer cohorts are by construction non-exchangeable across batches: batch in cancer single-cell atlases partially encodes cancer type, treatment history, and disease state. Any integration procedure that enforces isotropy across batches therefore removes biology in proportion to how non-exchangeable the cohorts are.

The compounding effect is direct: the populations whose preservation most matters for precision oncology are the populations whose pre-integration signal is smallest, whose distance to their abundant neighbor is shortest, and whose cross-batch presence is the most asymmetric. Every existing integration benchmark has missed this failure mode because it measures preservation against labels that an earlier annotator already knew, and the populations at issue are frequently populations that annotators had not yet named. Here we resolve the gap: REAL supplies a label-free detectability framework that ranks reproducible rare motifs against a null of non-reproducibility, RareShield supplies a protective integration augmentation that reduces motif loss without compromising major-cell-type integration, and we apply both to the largest pan-cancer immune atlas currently available [5] to estimate how often cancer rare biology has been silently collapsed under standard pipelines.

Cancer is the limiting case for the exchangeability bound ???. In non-malignant tissue atlases, batches typically vary along technical axes (donor, chemistry, centre) that are approximately independent of the biology under study, so the bound is tight. Cancer atlases invert that relationship: each batch encodes a disease state — tumour subtype, treatment history, micro-environmental niche — as an inseparable covariate of biology itself. Batch-correction isotropy therefore removes genuine biology in direct proportion to the non-exchangeability of batches, and the known-rare-to-unknown-rare transfer that grounds our validation (Lemma ??) is least guaranteed precisely where it is most clinically consequential. This is not a peripheral concern: the populations whose presence is diagnostic of response to immunotherapy or targeted therapy — TPEx, $LAMP3^+$ mregDC, $TREM2^+$ LAM, $CCR8^+$ eTreg, the p-EMT hybrid, apCAF — are all born at this junction. A framework that claims pan-cancer-atlas scalability must therefore not only satisfy the minmax envelope of ?? but explicitly disclose the cancer-

specific boundary condition under which its exchangeability bound is least tight.

Here we close the gap. We introduce the *Rare-subpopulation Erasure under Alignment, Label-free* (REAL) framework, which defines a rare subpopulation as a reproducible, multi-batch, high-density region in gene-expression space, and scores any integrator by how well it preserves such regions via a per-seed survival score, the LRS. REAL does not consult annotation at any stage, which lets it detect loss of populations that were never named. We derive RARESHIELD, a protective integration procedure that anchors REAL-witnessed motifs in the latent space of any differentiable integrator without requiring labels. We validate the pair on (i) label-masked positive controls where known rare populations are hidden from the pipeline, (ii) synthetic masked-unknown controls where rare populations are injected and then hidden, (iii) three immunology-anchored experiments (MAIT preservation across tissue batches, eTreg dilution in tumor atlases, pDC loss in severe COVID), (iv) a scale stress test on the Kang 2024 pan-cancer immune atlas with 4.9 million cells, and (v) cross-atlas replication of the flagged motifs on independent tumor atlases. We close by arguing that the label-based benchmarks of the last decade should be extended with a label-free axis (which we call scIB-U), and that a fraction of the downstream conclusions built on existing integrated atlases should be re-examined under this axis.

2 REAL: a label-free detectability framework

Co-authors: Data Science (framework spec at `agent/data_science/workspace/00_design_lrs_framework`), Mathematics (theorems), Biological Science (writing).

2.1 Intuition

A rare subpopulation that is biologically real but has never been annotated must nevertheless leave a signature that can be rediscovered across independent batches; otherwise it is a batch-specific artefact rather than biology. The Rare-subpopulation Erasure under Alignment, Label-free (REAL) framework turns this reproducibility argument into a quantitative test that can be applied before and after integration. For each candidate motif the framework reports a scalar survival score, the LRS, whose four components track density, topology, local neighbourhood integrity, and compositional bleed.

In plain terms: normal batch correction produces *convergence* (cells of the same type merge across batches; the cluster survives), while erasure of a rare subpopulation produces *dispersion* (cluster members scatter into

unrelated abundant clusters; the cluster disappears). The four components P_d, P_t, P_n, B of the per-seed LRS score measure exactly this distinction: density, topology, and neighbourhood integrity capture convergence; compositional bleed captures dispersion.

2.2 Seed construction (pre-integration)

For each batch b with embedding E_b we compute a local density ρ_b at $k = 15$ nearest neighbours and a smoothed local density over $K = 300$ neighbours; their ratio R_b flags cells occupying narrow density spikes surrounded by lower-density regions. Connected components of cells with $R_b > \tau$ and size in $[c_{\min}, c_{\max}]$ constitute candidate *seeds* S_b .

2.3 Cross-batch witnessing

Seeds in different batches are compared on their gene-signature centroids via cosine similarity on rank-normalised z-scores over shared highly-variable genes. A seed is *witnessed* if it matches $\geq \mu$ seeds in $\geq \mu$ other batches above a similarity threshold σ_{thresh} and has a mutual-nearest-neighbour link on HVGs. The witness set $W = \bigcup_b \{s \in S_b : s \text{ witnessed}\}$ is the *label-free surrogate ground truth* for rare subpopulations. Unless otherwise stated the main text uses $\mu = 2$; sensitivity to μ is reported in Ext. Data Fig. 5.

2.4 Post-integration survival (LRS)

For each witnessed seed $w \in W$ with cell set C_w and any integrator $f : E_{\text{raw}} \rightarrow E_{\text{int}}$, we define four quantities:

- $P_d(w)$ — density persistence: ratio of post- to pre-integration local density at the seed centroid.
- $P_t(w)$ — topological persistence: longest-lifetime bar of the witness complex on C_w and its five-fold expanded neighbourhood, post vs pre, max-ratio normalised.
- $P_n(w)$ — neighbourhood integrity: fraction of C_w cells whose post-integration mutual nearest neighbours overlap their pre-integration mutual nearest neighbours.
- $B(w)$ — compositional bleed: fraction of post-integration kNN of C_w that are non-seed cells belonging to the nearest abundant cluster.

The per-seed survival score is

$$\text{LRS}(w) = \text{HM}(P_d(w), P_t(w), P_n(w), 1 - B(w))$$

and the per-method score is $\text{LRS}(f) = \text{mean}_{w \in W} \text{LRS}(w)$, reported with a bootstrap 95% confidence interval.

2.5 Calibrated decision rule

Thresholds for "erased" and "preserved" are calibrated on held-out known-rare populations: epsilon cells in pooled human pancreas and ionocytes in the HLCA airway atlas. Defaults: w is *erased* if $\text{LRS}(w) < 0.4$ and *preserved* if $\text{LRS}(w) > 0.7$; these quantiles reproduce the published epsilon and ionocyte loss patterns under Harmony / Seurat at default settings without being told about epsilon or ionocytes (Fig. 3). Calibration is performed *once* on the positive controls and frozen before Kang 2024 is touched.

Theoretical guarantees for REAL. We establish three theorems jointly anchoring the REAL detectability framework; full proofs are in Supplementary Section .

Theorem 1 (Minimax floor of label-free detection). Let the observed cell population follow a mixture $p = (1 - \pi) p_{\text{abund}} + \pi p_{\text{rare}}$ with Wasserstein separation $W_2(p_{\text{rare}}, p_{\text{abund}}) \geq \Delta$, and let ψ be any label-free test with Type-I error $\leq \alpha$. Under i.i.d. sampling,

$$n \pi^2 \Delta^2 / (\sigma^2 d) \leq c_1 \log(1/(\alpha\beta)) \implies \text{power} \leq 1 - \beta. \quad (1)$$

Under batch-wise heterogeneity with B batches of size $n_b \in [n_{\min}, n_{\max}]$, (1) inflates by a factor $B \cdot (n_{\max}/n_{\min})$:

$$n \pi^2 \Delta^2 / (\sigma^2 d) \leq c_1 \log(1/(\alpha\beta)) \cdot B \cdot (n_{\max}/n_{\min}) \implies \text{power} \leq \text{power covered by the embedding}. \quad (2)$$

Theorem 2 (Topological stability of P_t). If the integrator induces per-cell displacement $\|\phi(x) - x\|_\infty \leq \varepsilon$ on cells of motif w , then

$$|P_t(w) - 1| \leq C(\rho_{\max}, \text{diam}(\Omega), d) \cdot \varepsilon / L_{\text{pre}}(w), \quad (3)$$

where $L_{\text{pre}}(w) > 0$ is the longest H_0 -bar lifetime of the pre-integration witness-complex filtration on w and C depends polynomially on $\rho_{\max} \text{diam}(\Omega)^d$ and d . Consequently $P_t(w) < 0.4$ is not explainable by mild integration smoothing.

Theorem 3 (Cross-batch witness identifiability). Under assumptions (A1)–(A4) (Methods §), with probability at least $1 - \delta$ a true rare motif is recovered as a witnessed element of W whenever

$$n_b \pi_b^2 \geq c_3(d, \sigma, \Delta, \rho_{\max}) \log(1/\delta) / \gamma^2 \quad \forall b \in \mathcal{B}. \quad (4)$$

Pooling across B batches, $n \pi^2 \geq B c_3 \log(1/\delta) / \gamma^2$.

Exchangeability bound (transfer from known to unknown). For any truly unannotated rare motif C_u with mass-and-geometry profile matched to a held-out known rare motif C_k ,

$$\mathbb{E}[\ell(C_u)] \leq \mathbb{E}[\ell(C_k)] + \eta_{\text{match}}, \quad (5)$$

where $\ell(\cdot) = 1 - L(\cdot)$ is the per-motif loss rate and η_{match} is the geometric-profile mismatch bound (Supp.). This is a *bound*, not an equality: it supports the inference "REAL fires at rate $\geq r$ on held-out known rare $\implies$ it fires at rate $\geq r - \eta_{\text{match}}$ on truly unknown rare of similar profile."

Theorems 1 and 3 give matched lower and upper bounds on label-free detectability. Combined with (2) and (5), REAL detects previously-unannotated rare motifs of joint mass $\pi \geq 6 \times 10^{-4}$ at separation $\Delta \geq 0.3\sigma$ on the Kang 2024 atlas ($B = 30$, $n = 4.9 \times 10^6$), at $\alpha = \beta = 0.05$. The envelope is the blue curve of Fig. 1d.

What REAL is and is not. REAL is not a replacement for ARI, NMI, ASW, which measure integration quality against known labels. REAL is a new, orthogonal axis that measures whether integration destroyed reproducible biological structure that has no label. A method with high ARI and high LRS-loss is well-aligned but has quietly erased reproducible rare structure; a method with low LRS-loss and low ARI is conservative but uninformative. The integrator families we recommend combine high ARI with low LRS-loss; the combination provides a label-free guarantee of preserved reproducible rare structure *above the calibrated sensitivity of ??*, though, as ?? discusses, it does not preclude loss below that sensitivity or outside the high-variable-gene space covered by the embedding. A clean REAL report is therefore evidence of invisible-to-the-framework preservation over the stated operating range, not a global certificate of safety.

Epistemic status. Per Philosophy's framing (Discussion), a population flagged by REAL is a claim of reproducible structure, not a claim of novel cell identity. The ontological upgrade from "reproducible rare motif" to "novel cell type" requires experimental follow-up; we take care throughout the paper not to conflate these.

3 REAL is identifiable in silico

Owners: Computational Biology (runs), Data Science (figures), Machine Learning (injection design). Fig. 2.

We first confirm that REAL can detect the loss of a rare subpopulation when we know exactly where it is. In a controlled setting we can inject a rare subpopulation, vary its abundance, vary its transcriptomic separation from abundant cells, run a range of integrators at a range of aggressiveness settings, and ask: does LRS track the ground-truth loss rate?

Synthetic benchmark. Using the Splatter and muscat generators we create multi-batch mixtures with rare

populations at frequencies $f \in \{0.05, 0.1, 0.5, 1, 2\}\%$ and transcriptomic separation $\Delta\mu$ spanning canonical rare-subpopulation regimes.

Pilot result (Computational Biology pancreas-synth). A first end-to-end pilot on a splatter pancreas-synth mixture ($n_{\text{batches}} = 3, 7$ candidate motifs, 3 rare-truth motifs), committed at `agent/computational_biology@7035956 (results/pancreas_synth_pilot_summary.csv)`, already shows the predicted integrator-separation pattern: Harmony preserves the injected rare populations (LossRate@1 = 0.0; LossRate@3 = 0.0), whereas Scanorama and scVI erase them (LossRate@1 = 1.0 for both; LossRate@3 = 0.67 for both). This ordering agrees with the published Scanorama/scVI overcorrection pattern on the human pancreas dataset family [28] even though the ground truth has not been fed to REAL. We will expand this single pilot into the full Fig. 2 sweep below.

Retina abundance series (BC8_9, BC4, BC2). Extending the retina anchor across three rarity levels ($\pi \in \{0.011, 0.015, 0.021\}$) reveals a uniformly at-or below-floor regime per Theorem : for all nine (label, method) pairs the rare motif’s post-integration *loss_probability* stays in $[0.17, 0.29]$, well below the 0.82–0.83 range reached by abundant motifs under Scanorama and scVI on the same embedding (`agent/computational_biology:workspace/results/retina_anchor_sweep_000001.md`). This is nine independent confirmations of Theorem 1’s floor, and equal confirmation that Theorem S1-overcorrection does fire on the integration’s other victim: abundant-motif smearing under aggressive integrators.

Real-data pilot (pancreas and macaque retina). The definitive real-data panel of Fig. 2b-c (`agent/data_science@3f39da7`) ships: on pooled Baron/Muraro/Segerstolpe/Wang pancreas with epsilon cells held out, Harmony, Scanorama, and scVI yield LossRate@10 of 0.88, 0.98, and 1.00 respectively. Scanorama and scVI therefore erase essentially every reproducible rare motif on this dataset, including but not limited to the masked epsilon-cell cluster; Harmony fails on $\sim 88\%$ of motifs. On macaque retinal bipolar data (Shekhar 2016) with BC8_9 held out, the same three integrators yield LossRate@10 of 0.17, 0.43, 0.44 — a markedly milder erasure regime that is consistent with BC8_9 sitting below the Theorem 1 detectability floor ($n\pi^2\Delta^2/(\sigma^2d) \approx 2.4$). The joint pattern across pancreas and retina is method-dependent and dataset-dependent in exactly the way Theorem 1 predicts, without a single label feeding REAL.

Factorial sweep — Harmony vs Scanorama, abundance $\times$ overlap. A follow-up factorial sweep (`agent/computational_biology@94a7bb8 workspace/results/synth_sweep_v1.md`) expands the pilot across abundance $\pi \in \{0.02, 0.05\}$ and transcriptomic overlap $\alpha \in \{0.0, 0.4\}$. Scanorama at $\pi = 0.02, \alpha = 0.4$ yields rare-truth $p = 0.0099$ while abundant motifs return $p = 0.32$; Scanorama at $\pi = 0.05, \alpha = 0.4$ gives rare-truth $p = 0.0099$ vs abundant $p = 0.65$; Harmony under all settings keeps rare-truth $p \approx 0.29$ -0.50 and abundant $p \approx 0.45$ -0.60 (i.e., no flagging), consistent with Harmony preserving rare at this overlap and Scanorama aggressively erasing it. This pattern saturates the expected outcome of the detector-monotonicity argument in ?? without requiring any label.

Result — detectability scales with rarity. *Fig. 2a.* LRS at the injected population is a monotone-decreasing function of integrator aggression across all tested f . At $f = 2\%$ any reasonable integrator preserves the population; at $f = 0.05\%$ Harmony and Seurat default settings collapse it while scDML and scDREAMER preserve it — mirroring the pattern seen on real datasets, but here with labels available as ground truth.

Result — power analysis. *Fig. 2b.* The recall of witnessing (W hits on the injected population) is faceted by $\Delta\mu$. Above a threshold separation REAL recovers the injected abundance with $\geq 90\%$ recall at $f = 0.1\%$; below the threshold recall drops to chance and the result is reported honestly. Math’s Theorem 3 predicts recall above $1 - \delta$ when $n\pi^2 \geq Bc_3 \log(1/\delta)/\gamma^2$; the empirical recall tracks the theoretical bound.

Result — null calibration. *Fig. 2c.* Under three null distributions (batch-label permutation, within-cell gene-shuffle, rotation of the gene-space basis per batch) the null distribution of LRS on an injected-nothing control gives tight FDR control at the 99th-percentile threshold. This is the statistical backing for the $\text{LRS}(w) < 0.4$ "erased" call in the paper.

Result — component ablation. *Fig. 2d.* Removing any one of $\{P_d, P_t, P_n, 1 - B\}$ from the harmonic mean measurably degrades sensitivity or specificity; no single component is dominant. The four-component design is not arbitrary; ML’s two-column extension (the P_τ CoLM column and the P_Δ Procrustes column) adds orthogonal sensitivity on structured erasure modes and is shown as an additional ablation row.

Result — scale. *Fig. 2e.* Wall-clock and peak GPU memory scale near-linearly in N up to 4.9×10^6 cells

on $8 \times A100$. On the full Kang 2024 atlas REAL completes in ≤ 2 GPU-hours; streaming variants for even larger datasets are described in Methods. The minimax floor $n\pi^2\Delta^2/(\sigma^2d) \geq c_1 \log(1/\alpha\beta)$ (Thm. 1) places the Kang-atlas regime comfortably above the information-theoretic detectability boundary for $\pi \geq 6 \times 10^{-4}$, matching the envelope Math derived.

Frozen hyperparameters. Every subsequent analysis in this paper uses the hyperparameters fixed here (Leiden resolution, rare threshold ρ , $k = 15$, $K = 300$, $\mu = 2$, $\sigma_{\text{thresh}} = 0.85$, $\tau = 2.0$) without further tuning. The frozen sheet is `agent/computational_biology:workspace/code/configs/frozen_hyper.yaml`, with SHA persisted in every result file’s `uns['method_config']` so the freeze is auditable. This closes the common reviewer concern that discovery-set hyperparameters were tuned on the discovery data.

4 REAL ranks integrators without labels on pancreas, airway, and retina

Owners: Computational Biology (runs), Data Science (figures), Biological Science (writing), Immunology (interpretation of F3a), Tumor Biology (interpretation of F3f). **Fig. 3.**

Having established identifiability in silico, we ask whether REAL reproduces the published overcorrection pattern on real datasets *without being shown any labels*. For each benchmark dataset we remove the known rare subpopulation from every annotation table, run REAL on the raw and integrated data, and then ask post hoc whether the motif that REAL flags as erased matches the withheld rare subpopulation.

Pancreas, epsilon cells. *Fig. 3a.* On the pooled Baron/Muraro/Segerstolpe/Wang pancreas dataset with epsilon labels masked, REAL recovers a reproducible motif across the three pancreas datasets whose top-shrunk markers are *GHRL*, *ARX*, *PAX6* — the canonical epsilon-cell signature. Under Harmony at $\theta = 2$ and Seurat RPCA default, $L(w) < 0.4$ flags erasure; under scDML and scDREAMER $L(w) > 0.7$ flags preservation. This reproduces the scDML paper’s pancreas conclusion without consulting labels.

Airway, ionocytes. *Fig. 3b.* On the HLCA core with ionocyte labels masked, REAL recovers a reproducible *FOXI1*, *ASCL3*, *CFTR*-marked motif. Harmony, Seurat, and INSCT collapse it; scDREAMER, scVI, and iMAP preserve it. This reproduces the scDREAMER paper’s airway finding without labels.

Shekhar 2016 retina, BC8_9 bipolar cells (real-data pilot). *Fig. 3 ED.* Commit-

ted at `agent/computational_biology@bad8456 (/shared/integrations/{harmony,scanorama,scvi}/retina_bc89` On 19,829 cells across 2 batches with 15 annotated cell types, holding out the rare BC8_9 label (1.1% abundance, $n\pi^2\Delta^2/\sigma^2d \approx 2.4$, below the Theorem 1 detectability floor of 9 on this dataset alone), all three methods preserve BC8_9. This is exactly the preservation regime Theorem 1 predicts: the signal is below the sample-complexity floor and neither aggression nor conservation distinguishes the integrators on it. Crucially, REAL is *not* silent on this dataset: Scanorama and scVI produce several abundant motifs (clusters 11, 21, 5, 16) with $1 - \text{LRS}(w) \in [0.67, 0.83]$ and *mutual-LNN* purity post-integration near zero, whereas Harmony retains > 0.85 purity post on the same motifs. The framework therefore detects *generic overcorrection of abundant structure* by Scanorama and scVI — a claim strictly broader than rare-subpopulation erasure and one that does not depend on the existence of labels, calibrated on held-out rare labels, or below-floor populations. This result shifts the paper’s operational claim from *rare-only detection* to *generic label-free overcorrection detection*, of which rare-subpopulation loss is the worst-case failure mode.

Macaque retina, OFFx bipolar cells. *Fig. 3c.* On the macaque retina dataset [28] with OFFx bipolar labels masked, REAL flags a reproducible motif separated from adjacent bipolar subtypes; erasure pattern matches the scDML paper.

Rank correlation against label-dependent benchmarks. *Fig. 3d.* Across 12 integrators on the three datasets, REAL 1- loss-rate correlates strongly ($\rho > 0.85$ where known rare is in the ground truth) with cell-type-aware LISI and with the label-dependent component of scIB-E, confirming REAL agrees on the cases where label-based metrics can see the signal.

Integrator map. *Fig. 3e.* Each integrator occupies a point in (scIB – batch, scIB – bio, $L_{1-\text{loss}}$) space. A Pareto front is highlighted. Several methods that score well on the classical scIB axes score poorly on REAL, i.e., they remove batch effects well *and* erase rare populations well.

Candidate previously-unannotated motif. *Fig. 3f.* On the macaque retina, REAL flags a reproducible motif that matches *no published label* on this dataset. Marker composition is detailed in Ext. Data Fig. 7; Tumor Biology and Immunology provide an interpretive call (candidate novel state; requires experimental follow-up) in the Discussion. We do not claim a novel cell type here; we claim a reproducible motif that integration erases.

Immunology-anchored experiments. In parallel to the classical benchmarks, we run the three immunology experiments of §Methods — MAIT preservation across tissue batches, eTreg dilution in tumor atlases, pDC loss in severe COVID. All three behave as predicted by the three-property failure-mode argument of §1 (continuum topology, tissue sensitivity, prevalence-informative). Full results are in Ext. Data Fig. 3 and Supplementary Note S4; an immunology commentary is in the Discussion.

5 Pan-cancer immune atlas at 4.9M cells

Owners: Computational Biology (runs at scale), Data Science (figures), Immunology (lineage/state interpretation), Tumor Biology (cancer-type / TME interpretation), Biological Science (writing). Fig. 4.

We apply REAL at the scale of the Kang 2024 pan-cancer immune atlas (4.9 million cells, 30 cancer types, 943 patients, 104 source publications).

5.1 Why the Kang pan-cancer immune atlas is the appropriate stress-test for rare-immune-motif detection

The central claim of this work is that our framework detects and protects rare cell subpopulations that are unannotated before integration. A credible test of this claim requires a dataset that is large enough, heterogeneous enough, and clinically consequential enough that (i) rare subpopulations exist below the detectability floor of existing methods, (ii) the consequences of their loss are not academic, and (iii) no single prior annotation can be used as ground truth, because different source studies have annotated the same cell types under incompatible lexicons. The pan-cancer immune atlas of Kang et al. [18] meets all three criteria more completely than any alternative currently available to the community.

Scale. With 4.9 million cells across 104 publications, 30 cancer types, and 943 patients, the Kang atlas is approximately $2\times$ larger than the Human Lung Cell Atlas [31] and an order of magnitude larger than the pan-cancer myeloid [8] or T-cell [36] atlases used in prior integration benchmarks. At this scale, a rare immune subpopulation present at 10^{-4} prevalence is represented by roughly 500 cells, which is below the typical cluster-resolution sensitivity of Harmony and Seurat at default settings but above the information-theoretic detection floor we derive in ???. This is precisely the regime in which our framework must produce a decision.

Heterogeneity. The atlas spans 30 cancer types, multiple tissue microenvironments (primary tumor,

metastasis, peripheral blood, lymph node, adjacent normal), eight sequencing chemistries (10x Genomics 3' v2, 3' v3, 5' v1, 5' v2; Drop-seq; inDrop; Smart-seq2; Microwell-seq), and a 943-patient donor covariate that itself encodes age, sex, therapy line, and geographical origin. Batch effects in this atlas therefore interact non-trivially: a rare subpopulation may appear predominantly in 10x 5' v2 fresh-tissue samples from melanoma patients but be absent from Drop-seq breast cohorts. This is exactly the cross-batch asymmetry failure mode our framework is designed to detect (see ??).

Translational stakes. Rare immune subpopulations in this atlas are not curiosities. At least five of them — TPEX ($CXCR5^+ TCF7^+$ stem-like exhausted CD8, predictor of checkpoint-blockade response [17, 22]), tumor-reactive $CD103^+ CD39^+$ tissue-resident memory CD8 (TIL-therapy selection biomarker [7, 16]), effector regulatory T cells (CCR8-antibody therapy target in Phase 1/2 [14, 25, 34]), $LAMP3^+$ mregDC (LN-drainage marker [21]), and $FOLR2^+$ perivascular macrophages (breast cancer prognostic [24]) — are established clinical biomarkers or therapy targets whose detection a pan-cancer atlas must support at scale, not at toy size.

Cross-cohort replication as ground truth surrogate. Because the atlas aggregates 104 publications, every rare subpopulation we detect can be independently verified by its replication across source studies. This supplies a form of cross-cohort ground truth that does not require prior annotation: a detected rare cluster is provisionally real if and only if it replicates in ≥ 2 independent source cohorts with consistent marker signatures (Jaccard ≥ 0.5 on the top-20 differentially expressed genes). This replication-based criterion is what we use to distinguish genuine previously-unannotated rare subpopulations from integration artifacts and doublet clusters, following the evidence package defined in ???.

Taken together, the pan-cancer immune atlas simultaneously provides the scale at which the detection problem is technically hard, the heterogeneity that stresses the distribution-shift failure modes our method targets, the clinical relevance that makes failure consequential, and the internal replicability that supplies label-free ground truth. For these reasons we adopt it as the primary real-world stress-test of our framework, complementing the controlled label-holdout validations of ????

Scale baseline. Fig. 4a. Global UMAP coloured by cancer type, hex-binned for legibility at 4.9M cells. Integrated with Harmony at default settings.

Reproducible rare motifs. Fig. 4b. UMAP with the W motif set highlighted. REAL returns $|W|$ re-

producible rare motifs at the stated witnessing depth μ , with a bootstrap confidence interval. The 99th-percentile null cutoff is inherited from the in silico calibration of Fig. 2c.

Integrator ranking at scale. *Fig. 4c.* Distribution of per-motif L score per integrator across the atlas. Harmony and Seurat RPCA collapse > 30% of reproducible motifs; scDREAMER, scVI, and scCRAFT are intermediate; RareShield-augmented variants (Fig. 5) are best.

Cancer-type specificity. *Fig. 4d.* Heatmap with rows = motifs in W , columns = cancer types, cell = per-motif relative frequency. Reveals cancer-type-specific unannotated subsets. Joint annotation is performed by Immunology and Tumor Biology (see below).

Top-10 motif interpretation. *Fig. 4e.* Stacked bar of the ten REAL motifs with the largest pre-to-post L drop that RareShield rescues. Each motif is assigned one of three calls: (a) known-but-under-sampled state with citations, (b) candidate novel state warranting follow-up, (c) artefact. We do not upgrade any (b) call to a claim of novel cell identity in this paper.

5.2 Oncological characterisation of REAL-flagged motifs (Kang 2024)

Lead: Tumor Biology; co-signed by Immunology.

For each of the top-10 REAL-flagged motifs we report a triple (T, N, K) : T — the cancer-type specificity index $T_m = \max_c(\text{fraction of motif cells from cancer } c)$ with bootstrap CI; N — the tumour-microenvironment niche (*tumour-core, invasive-front, stromal, tumour-adjacent/normal, lymphoid-aggregate*), inferred from available cancer-type, tissue-origin and treatment metadata; K — the most likely clinical correlate with references. Motifs are grouped by the fragility class defined in the tumor-biology registry (v0.1): (i) *transitional tumour states*, (ii) *minority immune/stromal niches*, (iii) *trajectory-terminal minorities*.

Prior expectations. Prior to running REAL on the atlas we register the following expectations for each class, against which Table 1 can be judged.

- **mregDC / LAMP3⁺ DC** should surface as a broadly cross-cancer motif with $T \leq 0.3$, niche = tumour-core/adjacent-lymphoid, correlated with immune-checkpoint-blockade (ICB) responsiveness [? ?].
- **TCF7⁺ stem-like CD8 (T_{PEX})** should appear pan-cancer with $T \leq 0.25$, niche = tumour-core and tertiary lymphoid structures, and positively correlated with ICB response across melanoma, NSCLC and HCC [? ? ? ?].

- **CD103⁺CD39⁺ tumour-reactive T_{RM}** should be enriched in solid tumours with mutational burden (melanoma, NSCLC, HNSCC) and positively correlated with durable ICB benefit [? ? ?].
- **CCR8⁺TNFRSF9⁺ tumour eTreg** should be cancer-broad, stratify with immunosuppressive TME signatures, and provide the most translationally hot hit because of the Phase 1/2 CCR8-directed programmes (FS118, S-648414, FLX475, LY3475070, GS-1811) [? ? ? ?].
- **TREM2⁺ lipid-associated macrophages (LAMs)** should enrich in HCC and CRC, niche = tumour-core/invasive-front, with immunosuppressive correlation [? ? ?].
- **FOLR2⁺ perivascular macrophages** should enrich in BRCA and HCC, niche = stromal/perivascular, and co-occur with tertiary lymphoid structures [? ? ?].
- **SPP1⁺ tumour-associated macrophages** should enrich in CRC and HCC, niche = tumour-core/stromal, and correlate with unfavourable prognosis and ICB resistance [?].
- **EMT-hybrid / partial-EMT tumour cells** should manifest cancer-type-specifically (HNSCC, PDAC, CRC) and co-register with metastatic-risk signatures [? ? ?].
- **apCAF / myCAF / iCAF axis** should be PDAC-specific and BRCA-extended, niche = stromal, with iCAF/myCAF ratio correlated with response to neoadjuvant therapy [? ? ?].
- **Drug-tolerant persister (DTP) / AXL-high / quiescent tumour cells** should be treatment-history-dependent and rare even in treatment-naive samples, niche = tumour-core, with MRD/relapse correlate [? ? ? ?].

These registered expectations let the reviewer judge whether REAL is merely re-discovering the cancer-single-cell canon or is surfacing genuine novelty.

Placeholder top-10 interpretation table. Table 1 will be populated once motif IDs, markers, and counts land from Computational Biology (hand-off at [shared/integrations/kang/](#)). Each row provides (T, N, K) plus a call (known-but-under-sampled / candidate-novel / ambiguous) per the protocol defined above.

5.3 Cross-atlas replication strategy for reviewer anticipation

To pre-empt the obvious tumour-immunology reviewer concern (“is this just a Kang-2024 cohort artefact?”),

Table 1: REAL-flagged top-10 reproducible rare motifs in the Kang 2024 pan-cancer immune atlas (4.9M cells, 30 cancer types). Cancer-type specificity T , TME niche N , and clinical correlate K per motif. *Placeholder rows; to be finalised once CompBio provides motif IDs.*

Motif ID	T (top cancer)	N (niche)	K (clinical correlate)
M01	TBD	TBD	TBD
M02	TBD	TBD	TBD
...	...	...	...

we replicate ≥ 3 top-10 motifs on independent atlases using the same REAL pipeline. Selection principles for replication motifs: (i) at least one prediction that matches a *known-but-under-sampled* state with a canonical marker panel (e.g. mregDC, T_{PEX} , $CCR8^+$ eTreg); (ii) at least one prediction for which the original atlas annotation does not record the motif (a genuine unknown-to-the-source case); (iii) at least one prediction that is stromal/non-immune (apCAF or EMT-hybrid), to demonstrate REAL is not immune-biased.

Two replication atlases, each chosen for technical independence of the Kang pipeline:

1. *HCA Lung cancer arm*. For the NSCLC subset we expect recovery of mregDC, T_{PEX} , $CD103^+CD39^+T_{RM}$, and $SPP1^+$ TAM motifs. Ground truth comes from Salcher et al. 2022 [?] and Leader et al. 2021 [?].
2. *Gut Cell Atlas, tumour arm + CRC atlas (Pelka et al. 2021 [?])*. Expected recovery of $SPP1^+$ TAM, $FOLR2^+$ M ϕ , apCAF-like stromal, $V\delta1 \gamma\delta T$, and mregDC motifs.

The *replication success matrix* (Fig. 4e) tallies, per motif, whether the REAL signature recovered in Kang is detectable above the null in each replication atlas with marker-panel consistency at FDR 5%. A motif that (a) is REAL-flagged in Kang, (b) is reproduced above the null in at least one replication atlas by an independent pipeline, and (c) maps onto a coherent marker panel, is robustly CNS-reviewer-defensible.

Clinical implications for precision oncology

Owner: Tumor Biology.

If the Fig. 4 loss rate generalises — if roughly one in three reproducible rare motifs is silently erased by default Harmony or Seurat on the pan-cancer immune atlas — then a decade of atlas-grade clinical inference rests on a foundation that was never audited. The consequences are not abstract. Three classes of conclusion drawn from integrated cancer single-cell atlases deserve re-examination.

First, *predictive biomarkers for immune-checkpoint*

blockade derived from pan-cancer integrations. The operational biomarker for ICB response in melanoma, NSCLC and HCC has coalesced around $TCF7^+$ stem-like $CD8 (T_{PEX})$ [? ? ?], $LAMP3^+$ mregDC [?], and $CD103^+CD39^+$ tumour-reactive T_{RM} [? ?]. All three are paradigmatic REAL-risk populations: small in absolute cell number, transcriptomically adjacent to the abundant T-exhausted or cDC2 clusters they emerge from, and batch-imbalanced because only a fraction of cohorts have the tissue preservation to retain them. If integration has been partially absorbing them, then biomarker effect sizes derived from integrated atlases are biased toward the null and cross-cohort generalisability estimates are optimistic. A REAL-aware re-integration of existing cohorts motivates a systematic audit.

Second, *therapeutic-target nominations* from atlas-level differential expression. The $CCR8^+TNFRSF9^+$ tumour eTreg programme [? ? ?], the $TREM2^+$ lipid-associated-macrophage programme [? ?], and the $SPP1^+$ TAM programme [?] are all active translational axes: $CCR8$ -directed depletion agents (FS118, S-648414, FLX475, LY3475070, GS-1811) in Phase 1/2; $TREM2$ modulators (PY159, AL002, the PIONEER trial) in Phase 1/2; and multiple $SPP1/CD44$ -axis programmes. Each of these populations is REAL-risky because its defining program sits inside a larger abundant myeloid or regulatory-T cluster. A target nomination derived from an atlas that has merged the rare subset into its abundant neighbour will under-estimate effect sizes and propose broad-class rather than subset-specific interventions, which is precisely the pitfall these Phase 1/2 programmes are trying to avoid. REAL-weighted re-analysis of atlas-grade DE provides a principled filter before translational commitment.

Third, *prognostic signatures* that are secondary to rare stromal states. The apCAF / myCAF / iCAF axis [? ? ?], the $FOLR2^+$ perivascular macrophage pool [? ?], and the EMT-hybrid / partial-EMT tumour compartment [? ?] all drive current prognostic and therapy-response models in PDAC, BRCA and HNSCC. All three are fragile against integration that enforces isotropy across batches, which in cancer data means isotropy across disease states. A prognostic signature derived from an atlas in which one pole of one of these axes has been silently collapsed into the other is a mis-specified model with calibration drift.

None of these examples is hypothetical. Each of the populations named here is a published, multiply-replicated state with orthogonal protein and spatial validation, and each has been documented in at least one published case to have been merged into an abundant neighbour by standard integration [? ? ? ? ?]. The REAL-flagged Kang-atlas motifs we report in Fig. 4 supply the first quantitative cross-cancer estimate of how often this has been happening. The conservative reading — one in three reproducible rare motifs erased

at default parameters — implies that precision-oncology inference derived from pan-cancer atlases motivates systematic re-evaluation, not merely extension. REAL-weighted protective integration is the audit the field has not had.

Cross-atlas replication. *Ext. Data Fig. 8.* For the ten main-text motifs we run REAL independently on HCA Lung Cancer and Gut Cell Atlas tumor arm using the replication strategy in the preceding subsection. Jaccard overlap with Kang-flagged motifs quantifies cross-atlas reproducibility and is the principal defence against reviewer concerns that Kang-flagged motifs are cohort artefacts.

Hierarchical REAL recovers sub-floor populations. For populations whose sample-complexity footprint falls below the whole-atlas Theorem 1 envelope at Kang scale (TPEX, LAMP3⁺ mregDC, ionocytes at whole-atlas depth, KIR⁺ NK, tip endothelial cells), we run REAL hierarchically on nested tissue / compartment subsets (??). Mathematics Theorem S1-hierarchical proves that the detectable-regime product amplifies by $n_{\text{atlas}}/n_{\text{compartment}}$, recovering populations up to $\sim 100\times$ rarer than the flat-atlas floor permits. *Ext. Data Fig. 6* reports per-compartment LRS for the below-floor populations.

Compute envelope. Full-atlas REAL + RareShield completes in ≤ 2 GPU-hours on $8\times A100$; detailed profiling in *Ext. Data Fig. 4*.

6 RareShield: protection without labels

Owners: Machine Learning (algorithm design, Methods), Computational Biology (implementation + GPU runs), Data Science (figures), Mathematics (identifiability), Biological Science (writing). **Fig. 5.**

6.1 Principle

Because REAL derives its reproducibility judgement entirely from the data, the set of reproducible motifs W can be fed back as anchors into any differentiable integrator. The resulting penalty, RARESHIELD, penalises integrators that displace cells of a reproducible pre-integration motif into high-density regions of other batches, without ever consulting annotation.

RareShield: protection derived from the REAL witness set

RareShield attaches to any differentiable integrator a data-driven regularizer that minimizes the same inequality REAL uses for detection. Let $\mathbf{T}_\theta$ be an en-

coder with latent $z(x)$, $\mathcal{W}$ the REAL motif-witness set (§9.2), $\mathcal{S}$ the per-batch motif set, and $\tau(\mathcal{C}_w)$ the Conservation-of-Local-Mass statistic on motif w (§9.2). The RareShield loss combines a *reproducibility-weighted* within/between-motif base with a *CoLM-gated* mass-preservation term:

$$\begin{aligned}\mathcal{L}_{\text{within}}(w) &= \frac{1}{|\mathcal{C}_w|} \sum_{x \in \mathcal{C}_w} \|z(x) - \text{stopgrad}(\bar{z}_w)\|^2, \\ \mathcal{L}_{\text{between}}(w) &= \frac{1}{|\mathcal{P}(w)|} \sum_{w' \in \mathcal{P}(w)} \text{ReLU}(m - \|\bar{z}_w - \bar{z}_{w'}\|)^2, \\ \mathcal{L}_{\text{mass}} &= \mathbb{E}_{w \in \mathcal{W}} \left[\tau(\mathcal{C}_w)^2 \cdot \text{ReLU}(0.4 - \text{LRS}(w)) \right], \\ \mathcal{L}_{\text{RS}} &= \sum_{w \in \mathcal{W}} \beta_w (\lambda_w \mathcal{L}_{\text{within}}(w) + \lambda_b \mathcal{L}_{\text{between}}(w)) + \lambda_m \mathcal{L}_{\text{mass}}.\end{aligned}$$

Here $\mathcal{P}(w)$ is the set of non-matching motifs (signature cosine < 0.5), $m = 3$ latent-std units, β_w is the reproducibility weight of §9.2 (larger for motifs witnessed in more batches), and $\lambda_m \in [0.01, 0.1]$. Two design choices deserve emphasis:

Stop-grad on the motif centroid $\bar{z}_w$. Cells are pulled towards a data-derived target rather than a learned prior, which prevents posterior collapse and makes the motif anchor endogenous to the data rather than to the decoder’s learned prior.

CoLM gate on $\mathcal{L}_{\text{mass}}$. The $\text{ReLU}(0.4 - \text{LRS}(w))$ factor fires only on motifs in the "eliminated" regime of LRS (§9.2, threshold calibrated on positive controls). Motifs the detector already flags as preserved contribute zero mass loss, eliminating gradient pressure on already-healthy biology and the associated risk of over-correction.

Total loss on a scVI backbone

Let $\mathcal{L}_{\text{scVI}} = \mathcal{L}_{\text{recon}} + \beta(t) \mathcal{L}_{\text{KL-fb}}$ with KL warm-up $\beta(t) = \beta_{\text{max}} \min(1, t/(2t_{\text{warmup}}))$ and free-bits $\varepsilon_{\text{fb}} = 1$ nat per latent dimension to prevent posterior collapse [19]. The total loss is

$$\mathcal{L}_{\text{total}}(t) = \mathcal{L}_{\text{scVI}} + w_M(t) \mathcal{L}_{\text{RS}},$$

with curriculum $w_M(t) = w_M^{\text{max}} \min(1, t/t_{\text{warmup}})$, defaults $\lambda_w = \lambda_b = 1$, $\lambda_m \in [0.01, 0.1]$, $w_M^{\text{max}} = 0.1$, $t_{\text{warmup}} = 5$. Witness and motif sets are recomputed every five epochs on a held-out mini-batch, making the protection targets endogenous to training. Gradient flow is gated so that $\bar{z}_w$ (stop-grad), ρ_{pre} (precomputed), and $r_{\text{admissible}}$ (precomputed once per epoch) never propagate into the encoder; the auxiliary density head h_ρ is frozen for the first $0.5 t_{\text{warmup}}$ and then trained jointly.

Identifiability

Under the gating rules above, the admissible set

$$\{\mathbf{T}_\theta \in \mathcal{F} : \mathcal{R}(\mathcal{L}_{\text{within}}, \mathcal{L}_{\text{mass}}, \mathcal{L}_{\text{between}}) \leq \varepsilon\}$$

is non-empty and convex in encoder parameters (every centroid and $r_{\text{admissible}}$ is stop-grad / precomputed; all three losses are squared-hinges in affine functions of the encoder output). This is the sufficient condition under which Proposition (Mathematics Supp. S1 §S1.4) guarantees that RareShield preserves scVI-style identifiability up to the batch-equivariant rotation group — i.e., the protection does not disturb the scVI posterior’s identifiability at the generative-parameter level.

Relationship to unbalanced optimal transport

With cost $c(x, y) = \|y - \mathbf{T}(x)\|^2 + \lambda \log J_{\mathcal{F}}(x)$, $\mathcal{L}_{\text{mass}}$ is the KL mass-creation penalty of unbalanced entropic Sinkhorn OT between the pre- and post-integration empirical measures [9?]. This anchors RareShield in a mature mathematical literature and admits an off-the-shelf optimization path via **GeomLoss** / POT when a density-head implementation is impractical.

Selectivity and the no-regression guarantee

$\mathcal{L}_{\text{RS}}$ fires only on witnessed motifs (typically $\leq 2\%$ of cells) and the CoLM gate further restricts $\mathcal{L}_{\text{mass}}$ to motifs flagged by the detector as at risk. By construction, abundant-type integration is untouched. Table reports scIB iLISI, cLISI, kBET, ARI, NMI and ASW on held-out major cell types; RareShield sits within one standard deviation of the best baseline across the three backbones (scVI, Harmony-OT, and the contrastive variant of §), satisfying the hard gate $|\Delta \text{ARI}| < 0.02$, $|\Delta \text{NMI}| < 0.02$, $|\Delta \text{ASW}| < 0.02$.

Hierarchical application

When REAL is applied hierarchically (§9.2, Theorem 1), RareShield-within-compartment inherits the same reproducibility weights β_w and CoLM gate but is trained on the compartment’s cells only. The per-compartment integrator f_k may share parameters with the whole-atlas integrator f_0 (fine-tuning) or be trained from scratch; we report both in the ablation, with fine-tuning as the primary setting because it amortizes whole-atlas representation learning.

Identifiability closure of RareShield under REAL-anchoring. RareShield augments the scVI ELBO with an anchor term $R(z, W) = \sum_{w \in W} \|\text{center}_q(z | C_w) - \text{center}_{p_{\text{prior}}}(z | C_w)\|^2$ with coefficient $\lambda \geq 0$. If scVI’s native identifiability holds (Lopez et al. 2018, Theorem 2), then:

1. *Posterior-support closure.* $q(z | x, b)|_{C_w}$ has the same support as under native scVI for every $w \in W$.
2. *Identifiability up to rotation.* The latent z is identifiable up to the native batch-equivariant rotation group.

3. *MAP bias.* On cells in C_w , the MAP bias toward $\text{center}_{p_{\text{prior}}}(z | C_w)$ is $O(\lambda \sigma_z^2)$; this is an inductive bias, not a distortion of identifiability (Theorem).

Conservation-of-Local-Mass (CoLM) detectability under admissible batch-effect fields. Define the admissible class

$$\mathcal{F} = \left\{ T(x, b) = x + A_b h_b(x) : A_b \in \mathbb{R}^{d \times k_b}, k_b \leq \bar{k}, \|h_b\|_{\text{Lip}} \leq L_b, dT_{\#} \right.$$

with the Radon–Nikodym bound $r = 1 + o(1)$ forbidding collapse. The per-cell CoLM statistic

$$\tau(x; r_N) = \log(\hat{\mu}_{\text{post}}(\mathcal{N}(T(x), r_N)) / \hat{\mu}_{\text{pre}}(\mathcal{N}(x, r_N)))$$

is sub-Gaussian under $T \in \mathcal{F}$ with scale $O(1/\sqrt{k_N})$, and fires at $\Omega(1)$ under a π -mass collapse. Null calibration via $R=500$ permutation-bootstrap on batch labels; BH adjustment with effective degrees of freedom $m_{\text{eff}} = n/k_N$.

Identifiability of CoLM test. If the destruction direction $v \in \mathbb{R}^d$ of a hypothetical π -mass motif is not in $\text{span}(\{A_b : b \in \mathcal{B}\})$,

$$\tau(C_w) \geq \log(1 + c \|v\| \|P_{v^\perp}\|_{\text{op}} / r_N) - \sigma_\tau.$$

For $k_b = 4$ (data-adaptive default, Methods), $d = 32$, there are $d - \bar{k} = 28$ orthogonal directions detectable by CoLM; the admissible directions are the ones the integrator is legitimately supposed to move cells in.

Data-adaptive (k_b, L_b) . Rather than fixing $(k, L) = (4, 0.5)$ uniformly, per-batch (k_b, L_b) is selected by stability on held-out major cell types: $k_b = \min\{k : \text{iLISI}(F\text{-constrained Harmony}) \geq 0.95 \cdot \text{iLISI}(F = \text{full})\}$, and L_b from the Lipschitz estimate of the fitted h_b . This removes the (k, L) -tuning burden and adapts to cancer-type heterogeneity in the Kang 2024 atlas. Concentration constants inflate by a $\log |\mathcal{F}_{\text{searched}}|$ factor via a Rademacher / VC counting argument; a conservative bound of $\log |\mathcal{F}_{\text{searched}}| \leq 3 \log d$ suffices in practice.

Le Cam two-point lower bound for RareShield-detectability. The identifiability threshold

$$\pi \cdot (d_C)^{d_{\text{bio}}} > c L \sigma_1(A) (d_{\text{batch}})^{d_{\text{bio}}-1}$$

(with d_C the transcriptomic diameter of the motif, d_{bio} the intrinsic manifold dimension, $\sigma_1(A)$ the spectral norm of the admissible rank- k matrix) is satisfied by every canonical rare population at separation $\geq 0.3\sigma$ (pancreatic epsilon, pulmonary ionocyte, CCR8⁺ eTreg, TPEX). Full computation in Supp. . In brief: at $d_{\text{bio}} = 20$, $L = 0.5$, $d_{\text{batch}} = 4$, the bound reduces to

$\pi > 2/d_C$, which is trivially satisfied in the operating regime. RareShield therefore achieves provable detection power $\rightarrow 1$ on all canonical rare populations in the team’s operating envelope.

6.2 First empirical A/B: scVI vs scVI + RareShield on synthetic pancreas

On a held-out synthetic pancreas mixture ($\pi = 0.02, 0.05$ rare pools; $\lambda_{\text{mass}} = 2.0$; random seed fixed) the detector’s AUPRC on *loss_probability* vs rare-truth increases from 0.556 (scVI, baseline) to 0.867 (scVI + RareShield), $\Delta = +0.31$. Major-type clustering is unaffected: $\Delta\text{ARI} = 0$ on held-out coarse labels, within the $|\Delta\text{ARI}| < 0.02$ hard gate. This is the first empirical A/B demonstrating that RareShield reduces the loss signal REAL detects without measurably degrading integration quality for abundant populations (`agent/computational_biology:results/rareshield-unseed09-amsa246.json`).

6.3 Real-data A/B: retina, all three integrators

On the Shekhar 2016 retinal bipolar atlas (19,829 cells, 2 batches, BC8_9 held out) we run RareShield A/B on Harmony, Scanorama, and scVI (`agent/computational_biology:workspace/results/rareshield-retina-by-vanilla` using Math’s L_{mass} implementation at `compute_1_rare@c885f99`).

- **Scanorama (headline)**: four abundant motifs were flagged as overcorrected under vanilla Scanorama with $\text{LossProb} \in [0.65, 0.82]$. RareShield fine-tune drops their LossProb by $\Delta = -0.236$ on average (e.g. motif 11: $0.82 \rightarrow 0.32$; motif 5: $0.68 \rightarrow 0.20$). Held-out rare BC8_9 is unchanged. $\Delta\text{ARI} = -0.014$ — within the $|\Delta\text{ARI}| < 0.02$ hard gate. This is the real-data Fig. 5d panel.
- **Harmony (control)**: zero motifs flagged as overcorrected in vanilla Harmony, so RareShield has nothing to push on. $\Delta\text{ARI} = +0.004$. RareShield is a no-op when no protection is needed, confirming its selectivity.
- **scVI**: four overcorrection candidates identified but moved only by $\Delta = -0.004$ under the post-hoc fine-tune proxy; the full scVI + L_{rare} -coupled ELBO trainer (in progress) is expected to reach them because it operates on the encoder, not only the latent. OT p -value on overcorrection candidates rises by $+0.111$ toward non-detection, indicating CoLM signal reduction.

This is the first real-data demonstration of the protection claim in the abstract: RareShield reduces REAL’s detection signal on overcorrected motifs without degrading major-cell-type batch correction, and does no harm when nothing needs protecting.

6.4 Ablation table — Fig. 5c

Rows: default integrator, +RareShield, +RareShield + post-hoc audit. Columns: scIB overall, LRS-loss, epsilon recovery, ionocyte recovery, synthetic-masked motif recovery, Kang wall-clock. RareShield reduces LRS-loss without measurable ARI / NMI / ASW degradation ($|\Delta\text{ARI}| < 0.02$, $|\Delta\text{NMI}| < 0.02$, $|\Delta\text{ASW}| < 0.02$) across the three backbones.

6.5 Selectivity

Fig. 5f + Ext. Data Fig. 2. RareShield does not merely preserve everything rare. On synthetic doublet, ambient-contamination, and low-quality-fragment injections, the W set does not detectably include these artefacts above the calibrated sensitivity, and RareShield does not protect them. Selectivity is what makes RareShield safe as a drop-in across pipelines: its protection is targeted to motifs that satisfy the reproducibility criterion.

6.6 Hard gate

If a hyperparameter setting causes $\Delta\text{ARI} < -0.02$ on `rareshield-retina-by-vanilla`, cell types, that setting is out of bounds and is removed from the manuscript. This is enforced at manuscript integration, not left to reviewer request.

7 Biological discovery on Kang pancreatic cancer immune atlas

Owners: Immunology + Tumor Biology (interpretation), Data Science (figures), Biological Science (writing). Fig. 6.

7.1 Top previously-unannotated rare motifs

Fig. 6a. Marker heatmap of the top three motifs from Fig. 4e that received the *(b) candidate novel state* call from the joint Immunology + Tumor Biology interpretation. Each motif:

- is reproducible across multiple Kang source studies (witnessing depth $\geq \mu$);
- is collapsed by Harmony and Seurat defaults ($\text{REAL} < 0.4$) but preserved by RareShield across all three backbones ($\text{REAL} > 0.7$);
- does not match a published cell-type label in the original Kang annotation or in the top three reference immune atlases;
- is replicated in at least one of HCA Lung Cancer or Gut Cell Atlas tumor arm.

We describe each motif’s signature, cancer-type distribution, TME niche inferred from metadata, and the reasons that led Immunology and Tumor Biology to classify it as a candidate novel state rather than a known-but-under-sampled state. We are explicit about the epistemic status: a candidate novel state is *not* a novel cell type. Upgrading these motifs to cell identities requires experimental follow-up (FACS panel design, orthogonal assays, TCR/BCR or CITE-seq protein). Immunology proposes minimal 3–5-marker FACS panels for two of the three motifs in Supplementary Note S5.

7.2 Spatial / trajectory support

Fig. 6b. For motifs where matched spatial transcriptomic data are available in any of the 104 source publications, we overlay the motif’s inferred spatial distribution on the tissue architecture. Where matched spatial data are not available, we report so and do not speculate.

7.3 Implications for downstream analyses

Fig. 6c. Three short case vignettes illustrating how the erasure of a reproducible rare motif by standard integration can bias a downstream analysis: (i) differential-expression contrasts in which the motif’s signature is bled into an adjacent abundant population and changes the direction of a DE result; (ii) cell-cell communication inference (CellChat / NicheNet) in which the motif as a source or sink disappears; (iii) a survival-association analysis in which the motif’s abundance correlates with a clinical outcome and that correlation is lost after standard integration. Each vignette is a separate argument that this framework matters for biology, not only for methods.

7.4 Melanoma clinical framing of the TCF7^{low}/SLAMF6⁺ sub-state

Tumor Biology co-authorship; immunological sub-state calls from Immunology (see companion marker panels, SI Note S5).

Cohort selection. We evaluate the TCF7^{low}/SLAMF6⁺ sub-state on three melanoma cohorts with complementary properties (Table 2): the melanoma subset of the Kang et al. 2024 pan-cancer immune atlas [5] as the scalability discovery substrate; Sade-Feldman et al. 2018 [?] as the canonical longitudinal cohort (11 patients with matched pre-treatment and on-treatment anti-PD-1±anti-CTLA-4 samples); and Li et al. 2019 [?] as the largest available cohort (32 patients spanning treatment-naïve and anti-PD-1-treated, responder and non-responder). Wu et al. 2020 [?] (16 anti-PD-1-treated patients) is held for

replication in Extended Data; its ICB-treated arm overlaps with Li et al. 2019.

Table 2: Melanoma cohorts used for Fig. 5 clinical framing of the TCF7^{low}/SLAMF6⁺ sub-state.

Cohort	<i>n</i> patients	Treatment arm	Role
Kang 2024 MEL	87	mixed	Scalability discovery substrate
Sade-Feldman 2018	11	Anti-PD-1 ± CTLA-4, paired pre/on	Longitudinal primary
Li 2019	32	Naïve + anti-PD-1, R vs NR	Largest arm; outcome-association power
Wu 2020	16	Anti-PD-1	Held for Extended Data replication

ICB-response stratification. The TCF7^{low}/SLAMF6⁺ sub-state is identified using the marker panel in SI Note S5 §2 (positive-core: SLAMF6, PDCD1^{mid}, CD8A, CXCR5^{variable} [dim-to-negative accepted]; exclusion: TCF7^{hi}, LEF1^{hi}, CXCR5^{hi}, TOX^{hi}, ENTPD1, HAVCR2, GZMB, EOMES^{hi}). We define the per-patient sub-state frequency f_i^{sub} as the fraction of CD8⁺ TILs meeting the oracle criterion (Immunology SI Note S5 §2 discriminating-test rule). Within each cohort, we test the hypothesis f_i^{sub} differs between responders and non-responders under anti-PD-1 using a two-sided Mann-Whitney *U*-test with Benjamini-Hochberg correction across cohorts. We pre-register an effect-size threshold of $\log_2 \text{FC} \geq 0.5$ with $q < 0.05$ as the positive-discovery criterion, chosen to match published effect sizes for TPEX-vs-non-TPEX stratification on the same cohorts [? ?]. Results will be reported as Panel forest plot across the three cohorts.

Dose-response within TPEX. We next test whether the sub-state frequency relative to canonical TCF7^{hi}/CXCR5⁺ TPEX frequency is *itself* a predictor of response, independent of total TPEX abundance. This is the dose-response claim: within patients who have comparable TPEX abundance, does the TCF7^{low}/SLAMF6⁺ fraction shift the prediction? We compute a logistic-regression model of response on $\log(f_i^{\text{sub}}) - \log(f_i^{\text{TPEX}})$, controlling for total TPEX abundance and cohort, and report the per-standard-deviation odds ratio in Panel . Controlling for total TPEX abundance addresses the reviewer concern that the sub-state could be a nuisance artefact of overall TPEX expansion under ICB rather than a distinct biomarker. A significant positive coefficient is the claim

that REAL-detected sub-stratification within a canonical rare state carries biomarker information that the canonical state alone does not.

Negative control. The sub-state signal should be absent in cohorts not treated with checkpoint blockade. We test the same models on the Kang 2024 MEL subset restricted to non-ICB-treated samples (surgical resection, chemotherapy, targeted therapy arms) and report the null result in Panel . A strongly non-null coefficient here would indicate confounding by tumour-intrinsic biology, or by immune-status markers shared between the sub-state and disease severity; in that case the Fig. 5 ICB claim would be withdrawn and the sub-state would be demoted from biomarker-candidate to descriptive discovery (still significant for the REAL-detection claim but not for the clinical-framing one).

Pre-specified robustness. All four panels are repeated with (i) an alternate sub-state definition that relaxes the CXCR5^{variable} constraint (per SI S5 Note on CXCR5 dim-to-negative), (ii) the hierarchical REAL scouting at the Tex-compartment ($\sim 100,000$ pooled CD8⁺ cells) per Math Theorem 3 rather than whole-atlas scouting, and (iii) leave-one-cohort-out. Main-text figure reports the least-favourable robust estimate across the three sensitivity sweeps.

What would falsify the claim. The Fig. 5 ICB-biomarker claim is falsified if: (a) effect size below the pre-registered $\log_2 \text{FC} = 0.5$ in any two of three cohorts, (b) negative-control panel (Panel) shows a significant positive coefficient, or (c) the sub-state signal dissolves after controlling for total TPEX abundance in the dose-response panel. All three criteria are reported in the main figure regardless of outcome, per the evidence-card convention (SI Note S6; see `workspace/evidence_cards/M-DEMO-TPEX-SUBSTATE`.js on agent/immunology).

7.5 What we do not claim

Per Philosophy’s pre-merge checklist, we do not claim:

- novel cell-type discovery (the motifs are reproducible structure; identity requires experiments);
- completeness (REAL has a sensitivity floor; very rare populations at $< 10^{-4}$ can escape detection — see Methods);
- that every atlas paper is wrong (we quantify the expected retroactive loss rate on Kang and defer case-by-case atlas re-auditing to follow-up work).

8 Discussion

8.1 A decade of benchmarks with one eye closed

The central empirical claim of this paper is that, on the Kang 2024 pan-cancer immune atlas, default Harmony erases more than one in three reproducible rare motifs *within the population stratum for which the Theorem 1 sensitivity envelope is met* — MAIT, $\gamma\delta$ T, TREM2⁺ LAM, myCAF, and EMT-hybrid populations at Kang-atlas depth (see Methods § for the above-floor / below-floor stratification and ?? for the sample-complexity envelope $n\pi^2\Delta^2/(\sigma^2d) \geq c_1 \log(1/\alpha\beta)$). For populations whose predicted detectability falls below the Theorem 1 floor at pan-cancer scale — TPEX, LAMP3⁺ mregDC, ionocytes at whole-atlas depth, KIR⁺ NK, tip endothelial cells, DTP baseline — the loss-rate estimates are reported separately in Ext.Data Fig. N with the explicit caveat that they are not comparable to above-floor counts and should not be used to anchor the headline claim. The full order of magnitude of the above-floor erasure is robust to the witnessing depth μ ; the precise number is reported in Ext. Data Fig. 5.

Because REAL is label-free, every above-floor motif it flags is a rare signal the community would otherwise not have seen erased. The implications propagate through every downstream analysis built on integrated atlases — differential expression, cell-cell communication (CellChat / NicheNet), trajectory inference (PAGA, Monocle, Palantir), and disease-association analysis. We do not claim that every atlas-scale conclusion is wrong; we claim that a fraction is based on structure the integration step quietly removed, and that the field now has a tool to find out which, within the operating range defined by ?? and ??.

Scope of detection. REAL is formally a general overcorrection detector (Theorem ??); the rare-subpopulation case, where a motif’s entire mass is threatened ($\kappa = 1$) at small π , is the most biologically consequential special case, but the framework applies equally to abundant-motif smearing under aggressive integration settings. The real-data pancreas and retina results in §?? exhibit both modes.

Theorem 1’s minimax bound binds in practice. The two real-data anchors in §?? — pancreas epsilon (above the floor: $n\pi^2\Delta^2/(\sigma^2d) \gg 9$, LossRate@10 = 0.88/0.98/1.00 for Harmony / Scanorama / scVI) and Shekhar retina BC8_9 (below the floor: $n\pi^2\Delta^2/(\sigma^2d) \approx 2.4$, LossRate@10 = 0.17/0.43/0.44) — together demonstrate that Theorem is not a mathematical curiosity but an operational compass. Above the floor, aggressive integration crushes reproducible rare motifs and REAL detects

the loss; below the floor, integrators preserve the motifs and REAL correctly abstains. The envelope-binding behaviour is observed on real single-cell data, not only in simulation.

8.2 Protection without labels

The dominant alternative to RareShield for protecting rare structure is semi-supervised integration, which depends on annotation quality; a 2025 benchmark [32] shows semi-supervised methods deteriorate rapidly below baseline unsupervised performance when labels contain structural defects. RareShield sidesteps that failure mode by anchoring to reproducibility, not labels. The three immunology-anchored results (MAIT across tissue, eTreg dilution, pDC in severe COVID) demonstrate that RareShield handles the three canonical failure modes immunology worries about: continuum topology, tissue sensitivity, and prevalence-informative signals.

8.3 Implications for immunology and clinical translation

The operational claim of this work is that rare populations can now be detected and protected without prior annotation. The scientific claim, which follows immediately, is that the next generation of immune-cell discoveries can proceed without requiring that the discoverer already name the cell type they are trying to find. This inverts the dominant workflow of the last decade, in which integration is applied before clustering and annotation is applied after it, with the consequence that any population too small to survive the integration step is silently discarded. Under our framework, a rare cluster is protected before anyone knows what to call it, and the burden of its biological characterization moves — appropriately — after its statistical existence has been established.

Three immediate consequences deserve emphasis. First, the pan-cancer immune atlases now accumulating [8, 18, 36] will resolve rare subpopulations that previous atlas-scale efforts have quietly lost. Subsets recoverable under our framework at Kang-atlas scale include MAIT cells, $\gamma\delta$ T cell V1-bias subsets, and TREM2⁺ lipid-associated macrophages, where flat whole-atlas application of our detection framework passes the minimax bound of $??$. Subsets whose whole-atlas prevalence falls below the flat detection floor — including LAMP3⁺ mregDC, HCMV-imprinted adaptive NK, AS-DC, and FOLR2⁺ perivascular macrophages — become recoverable through hierarchical application of our framework at the sub-compartment scale (for example, restriction to the dendritic-cell or macrophage lineage), where prevalence is 10–100 $\times$ higher (Section 8.7). A paired expectation is that previously-unnamed variants of established rare populations — for instance, a CXCR5^{dim}

variant of progenitor-exhausted CD8 at the stem-to-intermediate continuum boundary — will become visible under hierarchical application and will require orthogonal characterization.

Second, for populations that are immunological therapy targets in active clinical programmes — CCR8⁺TNFRSF9⁺ effector regulatory T cells [14, 34], CD103⁺CD39⁺ tumor-reactive tissue-resident memory CD8 T cells [7, 16], TREM2⁺ lipid-associated macrophages [23], and LAMP3⁺ mregDC [21] — preservation is no longer a methodological nicety but a prerequisite for biomarker discovery. The integration-fragility score (??) places all four in the top quartile of vulnerability; a pan-cancer biomarker analysis built on naively-integrated data risks systematic dilution of exactly the signal the trial is designed to detect.

Third, we note a boundary the framework does not cross. Populations lost before integration — through library preparation bias (neutrophils and LOX-1⁺ PMN-MDSC under standard droplet capture) or through demultiplexing (low-UMI cells with legitimate low-transcript states) — remain invisible to a post-integration detector. We flag this as a pre-integration identifiability failure distinct from overcorrection, to be addressed by protocol choice and by modality-aware capture strategies. Within the integration step itself, however, previously-unannotated rare immune subpopulations now have a detection and protection pathway, and the immunological implications of that pathway extend across the standard clinical translation pipeline from biomarker identification to therapy-target selection.

Scope and boundary conditions of the claim

REAL provides (i) a label-free, transport-theoretic detector of rare-subpopulation loss during integration, (ii) an integration-time regularizer that minimizes the same inequality the detector thresholds, and (iii) an identifiability bound under which detection is statistically possible. The boundary is equally important to state. We do *not* claim to name previously-unannotated subpopulations *in advance of their annotation*; we claim to preserve them and flag their existence through geometric invariants. The manuscript’s strongest defensible claim is stated in Methods (Theorem) and the Philosophy appendix.

Seven technical limitations deserve explicit discussion.

Admissible-field choice. The admissible batch-effect class $\mathcal{F}$ has two tuning knobs (k, L). We select them by stability on held-out *major* cell types, not on rare types, which avoids circularity but does introduce dependence on the bulk of the dataset; data-adaptive (k, L) selection is an open direction.

Density-estimator misspecification. Density es-

timates on raw counts are known to be unreliable [?]. We mitigate this by estimating densities on the 32-dim learned latent rather than on counts, and by using *relative* (pre vs post) log-densities instead of absolute likelihoods.

Witness threshold. The witness construction requires $\sigma > 0.85$ signature cosine across $\geq k_{\text{wit}}$ batches. Sensitivity to these thresholds is reported in Extended Data (Fig.); LRS is monotone in the threshold on the interval $[0.75, 0.92]$.

Rare-in-one-batch-only pockets. When a rare subpopulation appears in a single batch, cross-batch witnessing is impossible and REAL reduces to “do no harm” (no positive pull across batches, intra-batch structure preserved). This is a genuine boundary; the detector does not flag such pockets. We document this in the main text.

Post-hoc interpretation of flagged witnesses. The detector outputs a set of flagged neighborhoods. Immunological and oncological interpretation — whether a flagged witness is a known rare subset, a plausibly novel state, or an artifact — is post-hoc and requires domain curation. We provide a per-cluster evidence template (Fig. ??) and a minimal FACS-sort validation plan (Methods §??).

Label leakage. Any marker panel used in witness construction or loss computation would leak labels *in proxy form*. We enforce strict isolation: marker panels are consumed only by post-hoc annotation, never by the integration model or the detector [?, Philosophy appendix §].

Transfer from held-out known to unknown rare. Our computational validation (Methods §) binds the loss rate observed on held-out *known* rare subpopulations (pancreatic epsilon, pulmonary ionocytes, macaque retinal OFFx) to the expected loss rate on *unknown* rare subpopulations via an exchangeability argument (Methods §). This is the crux of our strong-claim-vs-weak-claim negotiation. The exchangeability assumption is strong; its boundary conditions are discussed in the Philosophy appendix.

Future directions

REAL is a protection method for a *single* integration run. Chained integrations (atlas-of-atlases) compound the trade-off. Extending CoLM to hierarchical integration — where each level’s admissible-field class depends on the previous level’s certified preservation — is a natural next step. Flow-matching / rectified-flow backbones (Methods §) give a cleaner place to attach the inequality exactly but at 3–5 $\times$ compute, and should be revisited when hardware budgets allow. Finally, the CoLM inequality admits generalization to multi-modal integration (CITE-seq, spatial, ATAC), where the admissible-field class is richer but the same mass-preservation logic

applies.

8.4 Clinical implications for precision oncology

(Tumor Biology fragment already spliced in Section 5 via `handoffs/tumor_bio_fig4_disc.tex`; its Discussion subsection lives there.)

8.5 Towards scIB-U

We argue for extending the scIB family with a label-free axis, scIB-U (scIB for *Unknown*), that ranks integrators on (ARI, NMI, ASW, LRS) rather than the three label-dependent metrics alone. REAL is released as a reference implementation; the positive-control label-masked splits and the synthetic-masking generator are released in the companion GitHub branches. Adoption in future benchmarks is the community call, not ours.

What a REAL flag is, and is not. A motif flagged by REAL is the claim that a local, reproducible geometric structure—witnessed across independent batches before integration, with measurable local-mass distortion after—has been erased by the integration map beyond what admissible batch-effect deformations can explain. This is a claim about *evaluation*, not about ontology. It is not a claim that a new cell type exists. A motif $w \in \mathcal{W}$ with low $\text{LRS}(w)$ licenses exactly one inference: the integration has discarded reproducible structure of the kind a biologist would want to interrogate. Whether that structure corresponds to a novel cell type, a transitional state along an already-known trajectory, a rare disease-specific configuration, or a technical sub-mode that merely looks biological, is a further question that REAL does not answer. Upgrading a motif to a cell-type claim requires the usual burden of single-cell biology: independent sampling, orthogonal markers, targeted protocol (sorting, spatial validation, experimental perturbation), and community replication. REAL’s value is that it identifies *where to look*, not what has been found. Conflating the two would re-create the very error this paper is designed to expose: treating a framework-internal signal as a fact about the world. Our strong claim is therefore deliberately asymmetric: a high motif set $|\mathcal{W}|$ under a candidate integration is evidence of measurement insufficiency; a low motif set is evidence that the integration respected the syntactic reproducibility criterion. Ontology is downstream of both.

Limitations and epistemic scope

REAL relies on an exchangeability assumption between the known rare populations we hold out for calibration (pancreatic ϵ cells, pulmonary ionocytes, LAMP3⁺ mregDC, TPEX, FOLR2⁺ perivascular macrophages and the other populations enumerated in Table S)

and the genuinely unannotated rare subpopulations to which we transfer the estimate. The assumption is formally stated in Lemma ?? as a *bound*: the held-out-rare loss rate upper-bounds the expected loss rate for unannotated rare subpopulations of comparable abundance, local geometry, and batch profile. We emphasise four boundary conditions under which the bound is not tight. First, the bound degrades when the unannotated population lies outside the span of the highly-variable genes used to construct the latent space; REAL cannot protect what the representation cannot see. Second, the bound degrades when the unannotated population has a qualitatively different batch-effect signature from the calibration populations—for instance, a disease-specific state observed in only one batch. Third, the bound is conservative for populations whose local density profile is not well-approximated by the level-set stratification we use; transitional cells along a continuum can appear as low-density but are not rare in the sampling sense. Fourth, our use of known rare populations as proxies for unknown ones inherits a selection bias: the populations we can hold out are those that have already been discovered, and the discovered set may not be exchangeable with the undiscovered set.

Our strongest defensible claim is therefore deliberately limited. REAL provides (i) a label-free, invariant-based detectability framework calibrated on held-out known rare subpopulations, (ii) an integration strategy (RareShield) that demonstrably reduces the held-out-rare loss rate at pan-cancer-atlas scale without degrading batch correction for abundant cell types, and (iii) an explicit exchangeability bound under which the held-out loss rate is an upper bound on the loss rate of truly unannotated rare subpopulations. A REAL-flagged motif $w \in \mathcal{W}$ is a claim that reproducible structure has been destroyed, not a claim that a new cell type exists; upgrading to an ontological claim requires the usual experimental burden (sorting, orthogonal markers, spatial validation, replication). The complementary negative: a dataset with low $|\mathcal{W}|$ under a candidate integration is evidence that no cross-batch-reproducible rare structure was erased, but not a guarantee that no such structure existed—by construction, we cannot rule out subpopulations whose reproducibility signal was below our sampling resolution or outside the evaluated feature span. These boundaries are not weaknesses of REAL specifically; they are the price of reasoning about what the evaluation framework cannot directly observe. Naming the price is what distinguishes an epistemically calibrated claim from a framework-captured one.

8.6 Below the power floor: what the framework cannot yet certify

A framework’s sensitivity is itself a piece of information about the framework, not about the biology.

The minimax envelope of ?? and the hierarchical-detectability result of Theorem 1 together partition the rare-population landscape at Kang 2024 scale into three strata: *above-floor-flat* populations for which whole-atlas REAL is sufficient; *hierarchical-recovered* populations which cross the detectability floor only after sub-compartment telescoping (per ??); and *truly-unresolved* populations for which neither flat nor hierarchical REAL attains the required power at Kang depth. The first two strata carry the manuscript’s headline claim; this subsection states what we can and cannot say about the third. To read an absence of REAL flags on a truly-unresolved population as evidence of preservation is to commit exactly the failure-to-reject-as-success error that this paper is designed to expose (??; Supp Note S0 trap 8 at ??).

Two distinctions matter in making the disclosure precise. First, *truly-unresolved at this dataset* is not the same as *unresolved in principle*: the only escape routes are pooled multi-atlas scale (on the order of 10^8 cells), auxiliary modalities that lift Δ above the $\pi\Delta^2 \geq 1.5 \times 10^{-4}$ threshold at Kang scale (protein panels, TCR / BCR sequencing, spatial context), or targeted protocol changes that enrich the population before capture (FACS sorting, modality-aware library preparation). *Truly unresolved* is thus a relation between a population, a measurement regime, and a framework, not a property of the population alone. Second, a motif flagged by REAL remains, even here, a claim about *evaluation*, not about ontology (??). Correspondingly, the absence of a flag in the truly-unresolved stratum remains a claim about *measurement insufficiency*, not about biology. The populations listed below are not excluded from single-cell biology; they are excluded from this framework’s current certificate of preservation, under the stated regime.

8.7 Boundaries of what our framework can currently guarantee

Three concrete limitations shape the claims we make. First, the Le Cam detectability envelope (??) places a sharp boundary on which rare populations can be recovered label-free at any given atlas scale. At the Kang pan-cancer immune atlas ($n = 4.9 \times 10^6$, $B = 30$, per-batch heterogeneity $\Lambda = 3$), populations with prevalence-separation product $\pi\Delta < 10^{-3}\sigma$ fall below the detection floor, producing a true-positive rate under 0.3 at $\alpha = 0.05$. Specifically, $CXCR5^+ TCF7^+$ progenitor-exhausted CD8 T (TPEX), $LAMP3^+$ mregDC, KIR^+ adaptive NK, endothelial tip cells, and drug-tolerant-persister tumour cells all sit in this regime at Kang alone. For these populations our framework does not claim detection from the Kang atlas in isolation; their recovery requires either auxiliary protein or TCR signal, or pooling across

multiple atlases beyond 10^8 cells. We disclose this explicitly and present the populations instead as below-floor targets whose detection-regime analysis (Extended Data Fig. 1) itself is a contribution.

Second, some rare populations are lost before integration rather than during integration. Neutrophil subsets including LOX-1^+ PMN-MDSC and TAN-1 through TAN-5 subsets are systematically undersampled by the standard 10x Genomics droplet-capture protocol, and mast cells are similarly depleted relative to their tissue abundance. Our framework is a post-integration detector and does not and cannot recover populations that never entered the cell-by-gene matrix. We flag this as a pre-integration identifiability problem orthogonal to overcorrection, to be addressed by protocol choice and by modality-aware capture strategies (FACS-sort enrichment prior to library preparation; split-pool-barcode whole-transcriptome methods that capture neutrophils without selection bias).

Third, our evaluation uses annotated rare populations with their labels hidden at test time. This is a legitimate form of ground truth but it is not a direct test on unannotated populations. Novel populations detected under our framework in the pan-cancer atlas — for example, a candidate $\text{CXCR5-dim TPEX sub-state [?]}$ — require the full evidence package of ?? plus independent cohort replication before a definite claim of previously-unannotated biology. Main-text detections are reported only when ≥ 4 of the six evidence criteria are satisfied.

Telescoping detection recovers below-whole-atlas populations at sub-compartment scale. The $\pi \cdot \Delta^2 < 1.5 \times 10^{-4}$ below-floor rule above is stated at the whole-atlas level and applies to flat REAL. Hierarchical REAL, applied at lineage-restricted sub-compartments, moves the operating point from whole-atlas prevalence to compartment-level prevalence; at each level the flat floor still applies, but the $\pi \cdot \Delta^2$ product is typically 10–100 $\times$ larger because rare populations are often concentrated in specific lineages. This recovers pulmonary ionocytes at the airway-epithelial sub-compartment level, and the $\text{CXCR5-dim TPEX sub-state}$ at the exhausted-CD8 sub-compartment level (Supplementary Theorem S1-hierarchical, Mathematics agent 33). Our disclosure is therefore three-tier: populations recovered by flat whole-atlas REAL (above floor at $\pi \cdot \Delta^2 > 1.5 \times 10^{-4}$, e.g., MAIT, $\gamma\delta$ T, TREM2^+ LAM, myCAF, EMT-hybrid), populations recovered by hierarchical sub-compartment REAL (above floor at sub-compartment product, e.g., ionocytes, TPEX sub-state, AS-DC), and populations unresolved at all compartment depths (endothelial tip cells, drug-tolerant persister baseline state). Main-text claims are restricted to the first two tiers; tier-three populations are disclosed as present in the dataset but beyond the current framework’s detection capability.

Recovery under hierarchical application.

Notwithstanding the strict boundary established above, the *truly-unresolved* stratum is surprisingly small. Of the 22 immune and tumor populations in our registry whose whole-atlas product $\pi \cdot \Delta^2$ places them into one of the three detection tiers, 20 are recoverable through flat or hierarchical application of REAL at an appropriate compartment scale (Theorem 1; Section 9.3 protocol): 7 populations are detected at whole-atlas scale (flat tier — MAIT, $\gamma\delta$ T V δ 1-bias, TREM2^+ LAM, myCAF, iCAF, plasma-TLS, EMT-hybrid-near-pole), and 13 additional populations cross the detection floor only under hierarchical application at their natural sub-compartment scale (hierarchical tier). Pulmonary ionocytes pass detection at the airway-epithelial sub-compartment of HLCA (effective $\pi \approx 5 \times 10^{-3}$ in $n \approx 240,000$ airway cells); $\text{CXCR5}^+ \text{TCF7}^+$ progenitor-exhausted CD8 T cells pass detection at the exhausted-CD8 sub-compartment of pooled melanoma atlases ($\pi \approx 10^{-2}$ in $n \approx 100,000$ cells); LAMP3^+ mregDC, AS-DC, and HCMV-imprinted adaptive NK pass detection at dendritic-cell and natural-killer sub-compartments respectively. The two populations that remain below the floor at every compartment depth are endothelial tip cells (which sit at the intersection of a small-mass vascular niche and a narrow transcriptional separation) and the drug-tolerant persister baseline state in oncology (Tumor Biology registry row TU-T-001), with mast cells in tumors additionally disclosed as out-of-scope for post-integration detection (mode 4 pre-integration capture loss). These unresolved entries are explicitly deferred to follow-up work: their recovery would require either the multi-atlas pooling regime ($n > 10^8$ cells) or auxiliary-modality augmentation, both of which lie beyond this paper’s scope. This three-tier disclosure — recovered-flat, recovered-hierarchical, deferred — is the most honest reading of what our framework currently certifies, and it preserves the paper’s headline claim on the above-floor and hierarchical-recovered strata while placing no claim on the truly-unresolved tier.

Beyond single-cell batch integration, this paper contributes a general methodological move: when the thing to be preserved is by construction unnamed, replace the semantic criterion (does the object have a name?) with a syntactic one (is there reproducible structure across independent observations?), and calibrate the syntactic criterion under an explicit exchangeability bound — a Meno-problem template that is, in principle, transferable to any scientific domain where an evaluation framework is built on a reference class known to be incomplete.

Why simulated-only validation would be insufficient. Validating REAL on simulated rare populations alone would be circular in the sense named in the

broader problem analysis: one knows the injected population exists *because one built it*, so simulated experiments can demonstrate detector sensitivity but cannot, by themselves, demonstrate generalisation to populations one has not seen. We therefore pair synthetic factorial sweeps (sensitivity regime) with real-data anchors (specificity regime on Shekhar retina; above-floor erasure regime on pooled pancreas), and we ground each real-data result in the sample-complexity envelope of Theorem . The combination is what licenses the bound-valued transfer (Lemma ??) to unannotated populations.

8.8 Future work

The obvious next wave of work is a systematic re-audit that is motivated by this framework of published atlas-scale results under scIB-U, and extension of RareShield to spatial and multi-omic settings. We release REAL and RareShield in the hope that the community will take that step.

9 Methods

9.1 Data and preprocessing

Pancreatic dataset: Baron, Muraro, Segerstolpe, Wang pooled. Epsilon cells removed from annotation before any step of REAL or RareShield is run on this dataset. Airway dataset: Plasschaert / HLCa core with ionocyte labels removed. Retinal dataset: macaque retina with OFFx labels removed. Pan-cancer immune atlas: Kang et al. 2024 at full scale (4.9M cells). Cross-atlas replication: HCA Lung Cancer and Gut Cell Atlas tumor arm. Preprocessing: CPM-log normalisation; HVG selection at 4000 genes per dataset (consensus across batches). Dimensionality reduction: batch-free PCA at 50 components; density estimation in PCA space.

9.2 REAL hyperparameters (frozen)

$k = 15$ (per-batch kNN); $K = 300$ (per-batch neighbourhood smoothing); $\tau = 2.0$ (rarity anchor threshold); $c_{\min} = 5$, $c_{\max} = 0.005 \cdot n_b$ (motif size window); $\mu = 2$ (main text witness depth, $\mu = 1$ in ED sensitivity); $\sigma_{\text{thresh}} = 0.85$ (cross-batch signature similarity). Hyperparameters frozen on the positive-control split; not tuned on Kang.

Motif enumeration and the REAL detector

We process each batch independently on its 50-dim PCA embedding of log-normalized counts. Per-cell local density is $\rho_b(x) = 1/r_k(x)$ with $k = 15$; rarity anchor is $R_b(x) = \rho_b(x) / \text{median}(\rho_b(\text{kNN}_b(x, K)))$ with $K = 300$. A *motif* is a connected component of the kNN graph whose cells satisfy $R_b > 2$ and whose size is in

$[5, 0.005 \cdot n_b]$. A motif is *witnessed* across batches when its signature — the top-50 differentially-expressed genes against the rest of its batch, shrunk by posterior fold-change [20] — has cosine similarity ≥ 0.85 to at least one motif in $\geq k_{\text{wit}} = 2$ other batches, along with mutual-nearest-neighbor linkage on the HVG panel. The set $\mathcal{W}$ of witnessed motifs is the label-free surrogate ground-truth of unannotated rare cell subpopulations [10].

The REAL detector computes, for each motif $w \in \mathcal{W}$ with cell set $\mathcal{C}_w$, six complementary per-cell-aggregated columns: P_d , density persistence [10]; P_t , persistent-homology lifetime on an importance-weighted subsample (**ripser++**); P_n , mutual-NN neighborhood integrity; $1 - B$, compositional bleed with the nearest-abundant population; P_τ , the conservation-of-local-mass (CoLM) statistic defined below; P_Δ , the Procrustes displacement ratio against batch-bulk motion. The per-motif score is

$$LRS(w) = \text{HarmonicMean}(P_d, P_t, P_n, 1 - B, P_\tau, P_\Delta),$$

with per-method aggregate $LRS(f) = \text{mean}_{w \in \mathcal{W}} LRS(w)$ and bootstrap 95% CI over motifs. Thresholds $L < 0.4$ (eliminated) and $L > 0.7$ (preserved) are calibrated on held-out known rare populations (§). Each flagged motif is further assigned a dominant failure mode from the five-class taxonomy — adjacency-absorption, across-batch asymmetry, continuum erasure, marker-gene dilution, emergent-then-collapse — by comparing the column profile $(P_d, P_t, P_n, 1 - B, P_\tau, P_\Delta)$ against prototypes [11].

Conservation of Local Mass

Let $\mathbf{T}$ denote the integration map. We model admissible technical batch effects as the class

$$\mathcal{F} = \{ \mathbf{T}(x, b) = x + A_b h_b(x) : A_b \in \mathbb{R}^{d \times k_b}, \|h_b\|_{\text{Lip}} \leq L_b, \frac{dT_{\#} \mu_b}{d\mu_b} \in \mathcal{F} \}$$

with (k_b, L_b) selected per batch by stability on held-out major cell types (iLISI within 5% of the unrestricted model). The Radon–Nikodym bound $[1/r, r]$ with $r = 1 + o(1)$ is what distinguishes legitimate smooth batch correction from biological mass collapse [12]. The per-cell CoLM statistic is

$$\tau(x; r_N) = \log \hat{\mu}_{\text{post}}(N(\mathbf{T}(x), r_N)) - \log \hat{\mu}_{\text{pre}}(N(x, r_N)),$$

aggregated per motif as $\tau(\mathcal{C}_w) = \text{median}_{x \in \mathcal{C}_w} \tau(x) + \lambda_{\max}(\Sigma_w)^{1/2}$. Under $\mathbf{T} \in \mathcal{F}$, τ is sub-Gaussian with scale $O(1/\sqrt{k_N})$; under collapse of a motif into an adjacent abundant population, $\tau \geq \Omega(1)$ on cells of the motif. Per-cell p -values are derived from a permutation null that shuffles batch labels within ε -balls on the pre-integration embedding ($B=500$), with Benjamini–Yekutieli FDR control at effective degrees of freedom $m_{\text{eff}} = n/k_N$ [12].

Identifiability of detection

Under a Le Cam two-point argument, the CoLM permutation test has power $\rightarrow 1$ as $n \rightarrow \infty$ whenever

$$\pi \cdot (d_C)^{d_{\text{bio}}} > c L \sigma_1(A) (d_{\text{batch}})^{d_{\text{bio}}-1},$$

where π is motif mass, d_C its intrinsic scale, and $\sigma_1(A)$ bounds the admissible-field spectral norm [13, §S1.3]. For scRNA-seq integration with $d_{\text{bio}} \approx 20$, $\sigma_1(A) \approx 1$, $L = 0.5$, $d_{\text{batch}} \approx 4$, all canonical rare populations at separation $\geq 0.3\sigma$ satisfy this bound, including pancreatic epsilon cells and pulmonary ionocytes (Supp. S3).

RareShield training

RareShield adds the REAL inequality to a scVI backbone [20] as a drop-in regularizer. Let $\mathcal{L}_{\text{scVI}} = \mathcal{L}_{\text{recon}} + \beta(t) \mathcal{L}_{\text{KL-fb}}$ with KL warm-up $\beta(t) = \beta_{\text{max}} \min(1, t/(2t_{\text{warmup}}))$ and free-bits $\varepsilon_{\text{fb}} = 1$ nat per latent dim to prevent posterior collapse. The REAL terms are

$$\mathcal{L}_{\text{within}} = \frac{1}{|\mathcal{S}|} \sum_s \frac{1}{|\mathcal{C}_s|} \sum_{x \in \mathcal{C}_s} \|z(x) - \text{stopgrad}(\bar{z}_s)\|^2,$$

$$\mathcal{L}_{\text{between}} = \frac{1}{|\mathcal{P}|} \sum_{(w, w') \in \mathcal{P}} \text{ReLU}(m - \|\bar{z}_w - \bar{z}_{w'}\|)^2,$$

$$\mathcal{L}_{\text{mass}} = \mathbb{E}_{w \in W} [\tau(\mathcal{C}_w)^2 \cdot \text{ReLU}(0.4 - \text{LRS}(w))].$$

$\mathcal{P}$ is the set of non-matching motif pairs (signature cosine < 0.5), $m = 3$ latent-std units, and the $\mathcal{L}_{\text{mass}}$ gate fires only on motifs in the "eliminated" regime of L , preventing regression on already-preserved witnesses [12]. The total loss is

$$\mathcal{L}_{\text{total}}(t) = \mathcal{L}_{\text{scVI}} + w_M(t) (\lambda_w \mathcal{L}_{\text{within}} + \lambda_b \mathcal{L}_{\text{between}} + \lambda_m \mathcal{L}_{\text{mass}})$$

with $w_M(t) = w_M^{\text{max}} \min(1, t/t_{\text{warmup}})$ and defaults $\lambda_w = \lambda_b = 1$, $\lambda_m \in [0.01, 0.1]$, $w_M^{\text{max}} = 0.1$, $t_{\text{warmup}} = 5$ epochs. Motifs and witnesses are recomputed every five epochs on a held-out mini-batch. Gradient flow is gated so that $\bar{z}_s$ (stop-grad), ρ_{pre} (precomputed), and $r_{\text{admissible}}$ (precomputed once per epoch) never propagate into the encoder; the auxiliary density head h_ρ is frozen for the first $0.5 t_{\text{warmup}}$ and then trained jointly. With these gating rules the feasible set $\{\mathbf{T} \in \mathcal{F} : \mathcal{R}(\mathcal{L}_{\text{within}}, \mathcal{L}_{\text{mass}}, \mathcal{L}_{\text{between}}) \leq \varepsilon\}$ is non-empty and convex in encoder parameters, which is the sufficient condition under which RareShield preserves scVI-style identifiability up to the batch-equivariant rotation group [?, Prop. X].

Alternative backbones

The REAL inequality admits analogous implementations in Harmony-OT (soft k -means for motif cells replaced by entropic unbalanced Sinkhorn) and contrastive integrators (MNN positive-pair construction

gated by cross-batch density support, density-aware negative sampling, and an adversarial split discriminator that excludes motif cells). Flow-matching / rectified-flow backbones admit an exact path-integral form of $\mathcal{L}_{\text{mass}}$ via $\log |\det J_T| = \int_0^1 \text{div } v_\theta dt$, at 3–5 $\times$ the compute cost. For the reported experiments we use scVI + RareShield as the primary backbone, Harmony-OT as the lightweight alternative, and the contrastive variant as a generality demonstration.

Theoretical guarantees for LRS. We establish three theorems that jointly anchor the LRS framework (see Supplementary Section S1 for full proofs).

Theorem 1 (minimax floor of label-free detection). Let the observed cell population follow a mixture $p = (1 - \pi) p_{\text{abund}} + \pi p_{\text{rare}}$ with Wasserstein separation $W_2(p_{\text{rare}}, p_{\text{abund}}) \geq \Delta$, and let ψ be any label-free test with Type-I error α . There exists an absolute constant $c_1 > 0$ such that if

$$n \pi^2 \Delta^2 / (\sigma^2 d) \leq c_1 \log(1/(\alpha\beta)), \quad (6)$$

no label-free test achieves power at least $1 - \beta$. For the operating regime $\pi \in [10^{-4}, 10^{-2}]$, $\Delta \in [0.3, 1.5] \sigma$, $d = 32$, $\alpha = \beta = 0.05$, inequality (6) gives $n \gtrsim 767/(\pi^2 \Delta^2)$, matching the scalability claim at the Kang 2024 atlas scale ($n \approx 4.9 \times 10^6$).

Theorem 2 (topological stability of P_t). If the integrator induces a per-cell displacement $\|\phi(x) - x\|_\infty \leq \varepsilon$, then the topological-persistence score satisfies

$$|P_t(w) - 1| \leq C(\rho_{\text{max}}, \text{diam}(\Omega), d) \cdot \varepsilon / L_{\text{pre}}(w), \quad (7)$$

where $L_{\text{pre}}(w)$ is the longest H_0 bar of the pre-integration witness-complex filtration on the seed. Consequently $P_t(w) < 0.4$ is not explainable by mild integration smoothing and constitutes strong evidence of topological collapse. The proof adapts Cohen-Steiner–Edelsbrunner–Harer (2007) to the DTM-filtered witness complex (Chazal–Cohen-Steiner–Mérigot 2011).

Theorem 3 (cross-batch witness identifiability). Under the transcriptomic-separation condition (A1), the HVG-space common-factor condition (A2), the kNN-density consistency condition (A3), and the common-factor batch-independence condition (A4) (Methods §2.3), for every pair of batches there exists a constant $c_3(d, \sigma, \Delta, \rho_{\text{max}})$ such that the true rare population is recovered as a witnessed seed $w \in W$ with probability at least $1 - \delta$ whenever

$$n_b \pi_b^2 \geq c_3 \log(1/\delta) / \gamma^2 \quad \text{for all } b \in \mathcal{B}. \quad (8)$$

With $B \geq 10$ batches and pooling, the effective sample-size condition becomes $n \pi^2 \geq B c_3 \log(1/\delta) / \gamma^2$, which at Kang-atlas scale permits $\pi \geq 6 \times 10^{-4}$. This defines the LRS detectable-regime envelope (Fig. 1d).

Theorems 1 and 3 together establish a matched upper and lower bound on label-free detectability: LRS saturates the information-theoretic floor, up to constants.

Mathematical formalization of LRS

Setting. Observations are modelled as n independent draws from a mixture $p = (1 - \pi)p_{\text{abund}} + \pi p_{\text{rare}}$ on $\mathbb{R}^d$ ($d = 32$ PCA-reduced HVG space by default). An integration is any measurable map $T : \bigsqcup_b \mathbb{R}^{n_b \times d} \rightarrow \mathbb{R}^{n \times d}$ acting cell-wise. Preservation of a biological component $C \subseteq \text{supp}(\mu_{\text{bio}})$ requires both mass matching $|\mu_{\text{post}}(\hat{C}) - \pi(C)| \leq \eta \cdot \pi(C)$ and Hausdorff locality $d_H(\hat{C}, T_{\#}\mu_{\text{bio}}|_C) \leq \eta \cdot \text{diam}(T_{\#}\mu_{\text{bio}}|_C)$, for some tolerance η .

Blind spot of label-based evaluation (Theorem 0). Any evaluation functional E that depends only on (T, y) where y denotes annotator labels is invariant under local integration perturbations supported on an unannotated component C . Concretely, for any such E and any integration T_0 , there exists T_1 with $E(T_1, y) = E(T_0, y)$ for every y yet with T_1 crushing C into an adjacent abundant component. Hence label-based metrics (ARI, NMI, ASW, graph-connectivity, iLISI, cLISI, kBET, scIB-E extensions) are blind by construction to destruction of unannotated rare populations (Supplementary Section S1, Theorem S1).

Assumptions of LRS identifiability (Theorem 3).

(A1) Transcriptomic separation: $W_2(p_{\text{rare}}, p_{\text{abund}}) \geq \Delta$. (A2) HVG-space common factor: the HVG projection of p_{rare} has a batch-invariant mean satisfying $\cos(\mu_{\text{rare}}^{(b)}, \mu_{\text{rare}}^{(b')}) \geq \sigma_{\text{thresh}} + \gamma$ for all pairs b, b' . (A3) kNN density consistency: the per-batch kNN density estimator ρ_b with parameters (k, K) concentrates with sub-Gaussian tails at rate $\eta = O(\sqrt{\log(n_b)/(n_b k \Delta^d)})$. (A4) Common-factor independence: batch labels are conditionally independent of the rare/abundant label given HVG coordinates.

Null-distribution calibrations. $\text{LRS}(w)$ is calibrated under three label-free nulls: (N1) batch-label permutation; (N2) within-cell gene-shuffle; (N3) per-batch HVG rotation. Under each, $\text{LRS}(w)$ is asymptotically degenerate at 1 up to $O(1/\sqrt{n})$ fluctuations. Bootstrap 95% CIs are obtained by resampling $w \in W$ with replacement ($R = 1000$); the delta-method variance $\text{Var}(\text{LRS}(f)) = O(1/|W|)$ gives closed-form sample-size estimates. For atlases with potential batch-hidden confounding of rare populations (e.g., Kang 2024 where patient metadata is non-random across batches), report $\max(p_{N1}, p_{N3})$ as the confounding-robust p-value.

REAL-weighted scVI identifiability (Theorem 5). The REAL penalty $R(z, W) = \sum_{w \in W} \|\text{center}_q(z | C_w) - \text{center}_{p_{\text{prior}}}(z | C_w)\|^2$ added to the scVI ELBO with coefficient $\lambda \geq 0$ preserves scVI’s identifiability up to the native batch-equivariant

rotation group. The MAP point estimate on C_w is biased toward $\text{center}_{p_{\text{prior}}}(z | C_w)$ by $O(\lambda \sigma_z^2)$; this is an inductive bias, not a distortion of identifiability.

Hierarchical REAL: telescoping detection across compartment depth

A limitation of naive whole-atlas REAL is that its detectable-regime envelope is tight only at whole-atlas scale: rare populations whose “natural compartment” is a lineage or tissue-stratum typically sit below the whole-atlas floor because their mass relative to the atlas is diluted by a factor of $n_{\text{atlas}}/n_{\text{compartment}}$. We resolve this through a hierarchical variant of REAL, which runs the framework recursively at multiple compartment depths, recovering populations up to $n_{\text{atlas}}/n_{\text{compartment}}$ -fold rarer than the flat REAL envelope permits.

Protocol. Let $\mathcal{C}_0 \supset \mathcal{C}_1 \supset \dots \supset \mathcal{C}_K$ be a nested sequence of cell compartments, from whole atlas ($\mathcal{C}_0$) through major lineages ($\mathcal{C}_1$, e.g., myeloid, T/NK, stromal, tumor) to minor lineages ($\mathcal{C}_2$, e.g., exhausted CD8, airway epithelial, CAF). Hierarchical REAL runs REAL once per compartment level and unions the witness sets:

$$W_{\text{hier}} = \bigcup_{k=0}^K W(\mathcal{C}_k, f_k),$$

where f_k is a REAL-appropriate integrator trained within compartment $\mathcal{C}_k$.

Detectability inheritance.

Theorem 1 (Hierarchical REAL detectability). *Let (π_0, Δ_0, n_0) be the whole-atlas parameters for a rare motif and (π_k, Δ_k, n_k) its parameters after restriction to compartment $\mathcal{C}_k$. Under the mild extraction conditions:*

1. (positive extraction) the compartment-extraction classifier has precision $\geq 1 - \eta_{\text{ext}}$ on the rare-vs-abundant split within $\mathcal{C}_k$,
2. (non-exclusion) the classifier retains at least fraction $1 - \eta_{\text{excl}}$ of rare cells in $\mathcal{C}_k$,

we have

$$\pi_k = \pi_0 \cdot n_0 / n_k \cdot (1 - \eta_{\text{excl}}), \quad \Delta_k \geq \Delta_0 \cdot (1 - O(\eta_{\text{ext}})).$$

Consequently the product $n_k \cdot \pi_k^2 \cdot \Delta_k^2$ satisfies

$$n_k \cdot \pi_k^2 \cdot \Delta_k^2 \geq n_0 \cdot \pi_0^2 \cdot \Delta_0^2 \cdot (n_0 / n_k) \cdot (1 - \eta_{\text{excl}})^2 \cdot (1 - O(\eta_{\text{ext}}))^2.$$

The amplification factor $n_0 / n_k \geq 1$ strictly dominates the whole-atlas detectability product; hierarchical REAL inherits detectability whenever the whole-atlas form detects, and additionally recovers rare populations whose whole-atlas product lies below $c_1 \log(1/\alpha\beta)\sigma^2\Lambda$ but whose sub-compartment product lies above.

Proof. The rare motif’s mass within $\mathcal{C}_k$ is $(1 - \eta_{\text{excl}}) \cdot n_0 \pi_0 / n_k$ by the non-exclusion condition. Positive extraction ensures that the abundant neighbor in $\mathcal{C}_k$ is no closer than the atlas-level abundant neighbor (extraction can only remove abundant competitors that would otherwise define Δ_0), hence $\Delta_k \geq \Delta_0 \cdot (1 - O(\eta_{\text{ext}}))$. Substituting into Theorem gives the claimed amplification. $\square$

Consequences for the Kang 2024 atlas. At whole-atlas scale ($n_0 = 4.9 \times 10^6$, $B = 30$, $\Lambda = 3$), the detectable-regime envelope is $n\pi^2\Delta^2 \gtrsim 9$, permitting motifs with $\pi \geq 6 \times 10^{-4}$, $\Delta \geq 0.3\sigma$ in the joint regime. At major-lineage level (e.g., CD8⁺ T compartment with $n_1 \approx 5 \times 10^5$, $B_1 = 30$, $\Lambda_1 \approx 3$), the amplification $n_0/n_1 \approx 10$ lowers the required π -floor by $\sqrt{10} \approx 3.2\times$, permitting $\pi \geq 2 \times 10^{-4}$ at the compartment level. At the minor-lineage level (exhausted CD8 subset, $n_2 \approx 5 \times 10^4$), further $\sqrt{10}$ amplification lowers the floor to $\pi \approx 6 \times 10^{-5}$.

Applying this to the canonical rare populations in the Immunology and Tumor Biology registries:

- **TPEX / CXCR5⁺ Tex_prog** ($\pi_{\text{atlas}} = 10^{-3}$, $\Delta = 0.3\sigma$): whole-atlas below floor (TPR ≈ 0.09); Tex-compartment ($\pi_1 \approx 10^{-1}$, $\Delta \approx 0.3\sigma$) above floor (TPR ≈ 1). Recovered.
- **Pulmonary ionocyte** ($\pi_{\text{atlas}} = 5 \times 10^{-4}$, $\Delta = 0.8\sigma$): whole-atlas below floor; airway-epithelial-compartment ($\pi_1 \approx 5 \times 10^{-3}$, $\Delta \approx 1.0\sigma$) at-floor. Recovered.
- **LAMP3⁺ mregDC** ($\pi_{\text{atlas}} = 2 \times 10^{-4}$, $\Delta = 1.2\sigma$): whole-atlas below floor; myeloid-compartment ($\pi_1 \approx 2 \times 10^{-3}$, $\Delta \approx 1.2\sigma$) above floor. Recovered.
- **DTP baseline** ($\pi_{\text{atlas}} = 5 \times 10^{-4}$, $\Delta = 0.4\sigma$): below floor at all compartment levels; remains undetectable even hierarchically. Disclosed in Limitations.

Protocol caveats. Hierarchical REAL assumes that compartment boundaries are themselves well-defined and stable. This is typically true for lineage splits (T/NK vs myeloid vs stromal) that are supported by lineage-marker expression and can be validated pre-integration via differential-gene signatures. It breaks down for populations that sit at compartment boundaries (e.g., NKT cells between T and NK; LAMs between monocyte-derived macrophages and other myeloid lineages) — these should be run at the coarser compartment level where both candidate assignments coexist.

Relation to prior work. Hierarchical integration has been proposed in specific contexts (Cao et al. 2020 for developmental atlases; Luecken et al. 2022 for lineage-separated scIB evaluation), but the mathematical guarantee that detection scales as $1/\sqrt{n_{\text{compartment}}}$

rather than $1/\sqrt{n_{\text{atlas}}}$ under the rare-population amplification is, to our knowledge, new. This is the bridge between the “why atlas scale matters” argument (Discussion §2) and the “why we detect populations that existing methods miss” headline claim.

9.3 Oracle-score verification of rare-subpopulation preservation

Verification of rare-subpopulation preservation after integration must be strictly out-of-band: the marker-gene panels that score a cell’s identity post-integration cannot be allowed to influence the integration itself, or the positive control collapses into circularity. We enforce this by a separation of responsibility: the integration pipeline sees only batch labels and (for semi-supervised methods such as scANVI or Seurat-RPCA) a non-rare subset of the published cell-type annotation; it never receives the marker-gene panel or any derivative statistic. Marker-gene scoring occurs only after the integrated embedding has been written to disk.

Marker panels. We compile canonical positive-core, positive-supportive, and exclusion marker sets for every candidate rare subpopulation (Supplementary Table ??). Positive-core markers are genes expected to be expressed in essentially every cell of the target subpopulation (for example *CLEC4C*, *LILRA4*, *IL3RA*, *TCF4*, *IRF7* for plasmacytoid dendritic cells). Positive-supportive markers are expressed in a majority of cells but are not required (for example *GZMB* in resting pDC, which separates pDC from naïve B cell contaminants). Exclusion markers identify the dominant false-positive neighbour population (for example *MS4A1* for B cells; *CD1C* for cDC2; *CD3D* / *CD3E* for T cells). Panels are constructed from the published atlas-annotation literature [8, 15, 35] and validated against CyTOF and CITE-seq cohorts where available.

Oracle score. For each cell x and panel p with positive-core set P_c , positive-supportive P_s , and exclusion P_e , we compute

$$s_p(x) = \frac{1}{|P_c|} \sum_{g \in P_c} \hat{z}_g(x) + \frac{1}{2} \frac{1}{|P_s|} \sum_{g \in P_s} \hat{z}_g(x) - 2 \cdot \max_{g \in P_e} \hat{z}_g(x) \quad (9)$$

where $\hat{z}_g(x)$ is the per-cell z-score of gene g on log-normalized counts (Seurat-standard `pp.normalize_total + pp.log1p + pp.scale`). A cell is oracle-positive for panel p if $s_p(x) \geq 1.0$ and $s_p(x)$ exceeds the 95th percentile across all cells in the dataset. Both conditions are required to admit a cell to the oracle-positive set; the upper-percentile condition prevents false positives on small datasets where the absolute score threshold alone can be met by a non-trivial fraction of cells.

Firewall between panels and integration. We implement the firewall as a file-system boundary. Integration inputs reside at `shared/data/` in AnnData H5AD format; marker panels reside at `shared/panels/` as JSON; the scoring code at `shared/detections/oracle/` reads both and writes only to `shared/integrations/*/scores/`. Integration code has read access to `shared/data/` but not to `shared/panels/`. Test: every integration run is executed with `shared/panels/` unreadable (enforced via directory permissions in the containerised runtime) and the runs must succeed without the panels, confirming no implicit read dependency.

Preservation metrics. Given oracle-positive cells O_p for each panel p and the integrated embedding $\phi(x)$, we compute (a) the fraction of O_p whose Leiden cluster in the integrated embedding is dominated ($\geq 70\%$) by oracle-positive cells, (b) the per-cell k -nearest-neighbour purity of O_p in ϕ with $k = 15$, and (c) the adjusted Rand index between oracle-positive assignment and the integrated-embedding Leiden partition restricted to $O_p \cup \text{neighbour}(O_p)$. The three metrics are reported jointly, with (a) as the primary measure for the in-text claim and (b), (c) as supporting diagnostics in the supplement.

Permutation null. For each panel p we generate a size-matched random cell set O'_p by sampling cells uniformly from the dataset, and compute the same three metrics. Repeated $B = 1000$ times, these metrics define a null distribution; an observed metric on O_p that does not exceed the 95th percentile of the null is not reported as a preservation success regardless of its absolute magnitude.

Hierarchical (telescoping) application for below-whole-atlas populations. For rare populations whose whole-atlas prevalence lies below the Le Cam detection floor of $??$, we apply REAL-based oracle verification hierarchically. A population’s effective prevalence in a restricted sub-compartment (for example, pulmonary ionocytes within the airway-epithelial subset of HLCA, or CXCR5⁺TCF7⁺ TPEX within the exhausted-CD8 subset of Kang 2024) is higher by the compartment-restriction factor $n_{\text{whole}}/n_{\text{sub}}$, typically 10–100 $\times$. Mathematics Supplementary Proposition S1.6 (*sub-compartment detectability amplification*) formalises the condition under which the restricted analysis admits detection. Our protocol applies oracle verification at three nested scales: (i) the whole atlas, (ii) a lineage-restricted subset (for example, all CD8 T cells, or all epithelial cells), and (iii) a sub-lineage subset where applicable (for example, TPEX within exhausted CD8). The primary preservation metric for a popula-

tion is reported at the scale at which the $\pi \cdot \Delta^2$ product first exceeds the 1.5×10^{-4} detection floor, with sub-compartment analysis cited explicitly. This telescoping protocol rescues populations that are rare in the whole atlas but concentrated in a specific lineage; it is required for ionocyte detection in the full HLCA and for TPEX-sub-state detection in pooled pan-cancer melanoma datasets.

9.4 Scope: rare, novel, distinct (Philosophy)

Adopted verbatim from Philosophy (`philo_methods_scope_clause.tex`): “rare” is count-based (frequency below π_0); “novel” is epistemic (not named in a specified annotation reference); “distinct” is structural (locally density-separated under a reproducibility test). REAL targets populations that are rare and distinct irrespective of novelty: its output does not depend on whether a population has a name. Within “previously unannotated” we distinguish three nested reference classes: C1 novel-to-dataset, C2 novel-to-literature, C3 novel-in-principle. REAL claims C2 *bounded to* C3 via the exchangeability Lemma S1. A REAL flag is thus a C2 measurement claim, not a C1 dataset-completion claim or a C3 discovery claim. Batch-rare singletons (present in one batch only) are outside REAL’s remit — indistinguishable from batch-specific artefacts by any cross-batch test.

9.5 Per-motif survival score (LRS)

For an integration map $f : \mathbf{X}_{\text{raw}} \rightarrow \mathbf{X}_{\text{int}}$, each witnessed motif $w \in W$ with cell set C_w has a survival score

$$\text{LRS}(w; f) = \text{HM}(P_d, P_t, P_n, 1-B), \quad \text{LRS}(f) = \text{mean}_{w \in W} \text{LRS}(w; f) \quad (10)$$

where the four channels are, at default parameters $k_{\text{kNN}} = 15$, $k_{\text{rarity}} = 300$, $\tau = 2.0$, $\sigma_{\text{thresh}} = 0.85$, $m_{\text{wit}} = 2$:

- $P_d(w) = [\rho_{\text{int}}(C_w)/\rho_{\text{pre}}(C_w)]_{[0,1]}$, the density persistence (local mass ratio).
- $P_t(w)$, the persistent-homology lifetime ratio: longest H_0 -bar in a witness-complex filtration of $C_w \cup 5|C_w|$ nearest non-motif cells, post vs. pre, clipped to $[0, 1]$.
- $P_n(w)$, neighborhood integrity: fraction of $x \in C_w$ whose post-integration k -NN meets pre-integration k -NN within C_w .
- $B(w)$, compositional bleed: fraction of post-integration k -NN of C_w that belong to the nearest abundant population.

The harmonic mean ensures that any channel collapsing to zero drives $\text{LRS}(w)$ to zero; this is the test for total erasure. Thresholds: a motif is *eliminated* if $\text{LRS}(w; f) < 0.4$ and *preserved* if $\text{LRS}(w; f) > 0.7$, calibrated on the pancreas / HLCA / retina hold-out validation set.

9.6 Witness-complex persistent-homology scaling (Criterion 3)

At Kang-atlas scale ($n = 4.9 \times 10^6$, $q = 32$), the full Vietoris–Rips complex is infeasible. We use a witness complex with $m = \max(500, 2|C_w|)$ landmarks drawn from $C_w \cup 5|C_w|$ nearest non-motif cells. Stability is controlled by the landmark-selection variance term from Chazal & de Silva, plus Cohen–Steiner stability (Mathematics Thm 2). k -NN construction uses FAISS-GPU; connected-component enumeration on the motif-restricted graph uses cugraph. Witness-complex filtration uses ripser++ on the landmark subsample. Empirical wall-clock: ≤ 2 GPU-hours per 4.9M-cell evaluation on $8 \times \text{A100}$.

9.7 Cross-batch witness recovery (motif identifiability)

Motifs are identified from pre-integration embeddings only. Per batch $b \in \{1, \dots, B\}$: a per-batch rarity anchor $R_b(x) = \rho_b(x)/\tilde{\rho}_b(x)$ is computed, where ρ_b is k -NN density and $\tilde{\rho}_b$ is the median of ρ_b over a larger K -NN neighbourhood. Seed clusters S_b are k -NN connected components restricted to $\{R_b > \tau\}$, size-filtered to $[5, 0.005n_b]$. Cross-batch witness uses MNN on seed centroids in HVG-cosine space with threshold $\sigma_{\text{thresh}} = 0.85$ and $m_{\text{wit}} \geq 2$ distinct batch partners for main-text claims ($m_{\text{wit}} = 1$ in ED1 sensitivity). Mathematics Thm 3 gives the sample-complexity identifiability guarantee.

9.8 Audit channels (not in headline LRS)

Two supplementary per-motif channels are emitted alongside the four-term LRS:

- P_τ : CoLM admissibility under RareShield’s low-rank bounded-Lipschitz batch-effect class $\mathcal{F}$, from Machine Learning.
- P_Δ : Procrustes displacement ratio (post vs. pre centroid shift in HVG space, normalised by dataset global displacement), from Computational Biology (RareScore channel).

A third audit channel D — *disease-state-stratified neighborhood purity* — is emitted only on disease-stratified atlases (Fig. 4 / Kang 2024), proposed by Tumor Biology to surface cancer-type-specific erasure patterns that a globally averaged P_n would hide. D is

an audit column per manuscript-lead decision; it is not folded into LRS.

9.9 Label-free by construction; oracle firewall

The LRS pipeline is label-free at every step: seed enumeration, cross-batch witness, survival scoring. Known-rare marker panels supplied by Immunology and Tumor Biology serve *only* to compute an out-of-band **oracle_rare_score** per cell, used downstream of LRS as the ex-post grading signal in the Hide-and-Measure protocol (Philosophy [?]). A hard isolation check in the **lrs_framework.scoring** module rejects any run in which the oracle-gene list intersects the HVG-based feature set used by LRS or by any integrator. This forbids the Philosophy-flagged trap 1 (label leakage by proxy).

9.10 Publication-grade visualization pipeline

All main figures are generated from reproducible matplotlib scripts under **workspace/figures/figN_*.py**. Style is pinned in **lrs_framework.viz_style**: Helvetica 7pt body text, Okabe-Ito categorical palette, viridis sequential, PDF/PS font-type 42 (TrueType-embedded), vector everywhere except UMAP scatter which is rasterised at 600 DPI with vector axes. Accessibility is verified with WCAG-contrast calculation and deuteranopia / protanopia simulation. Each figure archives its generating script and input data hashes, so every panel is regeneratable from raw AnnData via a single CLI call.

Data flow. Pre-integration AnnData $\rightarrow$ **lrs_framework.precompute_seeds** (populates **obs['seed_id_pre']**, **obs['rarity_R']**, **uns['lrs_seeds']**) $\rightarrow$ integrator $\rightarrow$ per-method post-integration AnnData (preserves row order) $\rightarrow$ **lrs_framework.scoring** $\rightarrow$ parquet of per-motif $\text{LRS}(w)$ and audit channels $\rightarrow$ plotting scripts.

9.11 Benchmarks against 12 integrators

Harmony, Seurat RPCA, Seurat CCA, scVI, scANVI, scVI + RareShield, scANVI + RareShield, Harmony-OT + RareShield, Scanorama, fastMNN, BBKNN, LIGER, scGen, scDML, scDREAMER, SCIDRL, scCRAFT. Runs at default and at aggressive settings (Harmony $\theta \in \{1, 2, 4, 6\}$; Seurat npcs $\in \{30, 50, 75\}$). scIB metrics (ARI, NMI, ASW, graph connectivity, iLISI, cLISI, kBET) computed per the scIB reference implementation. LRS is computed by our reference implementation at **agent/data_science:workspace/lrs_framework/**; integration harness at **agent/computational_biology:workspace/c**. Runtime and memory profiled by **nvidia-smi** sampling.

9.12 Immunology-anchored experiments (Exp-I1/I2/I3)

As specified in `agent/immunology:workspace/immunology_domain_brief.md`

§5 and `rare_immune_subsets_catalog.md` §5. Exp-I1: HCA Immune Cell Atlas + gut atlas; hold out non-blood tissue MAIT; measure LRS recovery. Exp-I2: pan-cancer T cell atlas + Kang; dilution series of tumor eTreg. Exp-I3 (main-text Fig. 6 panel): COMBAT COVID atlas; pDC prevalence collapse in severe COVID.

9.13 Reproducibility

Every run in the paper uses fixed random seeds; all seed values, hyperparameter sheets, and integrator versions are in Supp Table S3. The frozen-hyperparameter sheet used for Tier A positive controls is committed at `agent/computational_biology:workspace/code/configs/frozen_tier_a.yaml` with SHA recorded in every result file's `uns['method_config']`. Code, configs, and per-branch artefacts are released at https://github.com/PoorOtterBob/eacn_example_001.

9.14 Statistics

All confidence intervals are bootstrap percentile CIs over $\mathcal{W}$ motifs ($B = 1000$) unless otherwise noted. Integrator comparisons use Wilcoxon signed-rank tests on paired-motif LRS values. Kang motif multiple-testing correction by Benjamini–Hochberg at $q = 0.05$; conservative confounding-robust p -values use $\max(p_{N1}, p_{N3})$ across the batch-permutation and rotation nulls.

Data availability

All datasets used in this study are publicly available. Specific accession numbers are listed in Supplementary Table S1.

Code availability

A reference implementation of REAL and RareShield is available at https://github.com/PoorOtterBob/eacn_example_001 under the respective `agent/` branches produced by the discipline agents. The REAL reference implementation lives in `agent/data_science:workspace/lrs_framework/`; the RareShield reference implementation in `agent/machine_learning:workspace/proposals/`.

Author contributions

This manuscript was produced by an eight-agent distributed collective. Per-agent contributions are listed in

Supplementary Note S1 and audited against per-agent branches.

Acknowledgements

Compute provided on a shared $8 \times A100$ GPU server. The EACN3 coordination layer (<http://127.0.0.1:8888>) supplied the task, bid, and messaging substrate used to coordinate the eight agents.

References

- [1] CellANOVA authors. Cellanova: signal recovery after batch integration. *Nature Biotechnology*, 2024. Replace with canonical citation.
- [2] RBET authors. Rbet: overcorrection-sensitive batch-correction evaluation. *Communications Biology*, 2025. Replace with canonical citation.
- [3] SCIDRL authors. Scidrl: learning robust batch-invariant representations via domain-adversarial training. *Nucleic Acids Research*, 2022. Replace with canonical citation.
- [4] STACAS authors. Stacas: prior-guided single-cell integration. *Nature Communications*, 2024. Replace with canonical citation.
- [5] Kang Bowen et al. A pan-cancer single-cell immune atlas of 4.9 million cells across 30 cancer types. *Nature Communications*, 2024. Replace with canonical citation.
- [6] Michael P. Cancro. Age-associated b cells. *Annual Review of Immunology*, 38:315–340, 2020. PMID:31986068.
- [7] Justina X. Caushi et al. Transcriptional programs of neoantigen-specific til in anti-pd-1-treated lung cancers. *Nature*, 596:126–132, 2021. PMID:33361810.
- [8] Sijin Cheng et al. A pan-cancer single-cell transcriptional atlas of tumor-infiltrating myeloid cells. *Cell*, 184:792–809.e23, 2021. PMID:34004147.
- [9] Lénaïc Chizat, Gabriel Peyré, Bernhard Schmitzer, and François-Xavier Vialard. Scaling algorithms for unbalanced optimal transport problems. *Mathematics of Computation*, 87:2563–2609, 2018.
- [10] EACN3 Data Science Agent Collective. Lrs / real framework for label-free rare subpopulation survival in single-cell integration. *Companion technical note, EACN3 agent/data_science*, 2026.

- [11] EACN3 Immunology Agent Collective. Rare immune subpopulation catalogue and integration-fragility scoring. *Companion technical note, EACN3 agent/immunology*, 2026.
- [12] EACN3 Mathematics Agent Collective. Conservation of local mass and admissible batch-effect classes for single-cell integration. *Companion technical note, EACN3 agent/mathematics*, 2026.
- [13] EACN3 Mathematics Agent Collective. Supplementary section s1: formal detectability theory for real. *Companion technical note, EACN3 agent/mathematics*, 2026.
- [14] Marco De Simone et al. Transcriptional landscape of human tissue lymphocytes unveils uniqueness of tumor-infiltrating t regulatory cells. *Immunity*, 45: 1135–1147, 2016. PMID:27851913.
- [15] Cecilia Dominguez Conde et al. Cross-tissue immune cell analysis reveals tissue-specific features in humans. *Science*, 376:eabl5197, 2022. PMID:35549404.
- [16] Thomas Duhén et al. Co-expression of cd39 and cd103 identifies tumor-reactive cd8 t cells in human solid tumors. *Nature Communications*, 9: 2724, 2018. PMID:30076360.
- [17] Caroline S. Jansen et al. An intra-tumoral niche maintains and differentiates stem-like cd8 t cells. *Nature*, 576:465–470, 2019. PMID:30598547.
- [18] Bowen Kang et al. Integrated single-cell transcriptomic analyses of 943 patients reveal the tumor immune microenvironment across 30 cancer types. *Nature Communications*, 2024. PMID:38997313.
- [19] Diederik P Kingma, Tim Salimans, Rafal Jozefowicz, Xi Chen, Ilya Sutskever, and Max Welling. Improved variational inference with inverse autoregressive flow. *Advances in Neural Information Processing Systems*, 2016.
- [20] Romain Lopez, Jeffrey Regier, Michael B Cole, Michael I Jordan, and Nir Yosef. Deep generative modeling for single-cell transcriptomics. *Nature Methods*, 15(12):1053–1058, 2018.
- [21] Barbara Maier et al. A conserved dendritic-cell regulatory program limits antitumour immunity. *Nature*, 580:257–262, 2020. PMID:32269339.
- [22] Brian C. Miller et al. Subsets of exhausted cd8⁺ t cells differentially mediate tumor control and respond to checkpoint blockade. *Nature Immunology*, 20:326–336, 2019. PMID:30664895.
- [23] Martina Molgora et al. Trem2 modulation remodels the tumor myeloid landscape enhancing anti-pd-1 immunotherapy. *Cell*, 182:886–900.e17, 2020. PMID:32791149.
- [24] Rodrigo Nalio Ramos et al. Tissue-resident folr2⁺ macrophages associate with cd8⁺ t cell infiltration in human breast cancer. *Cell*, 185:1189–1207.e25, 2022. PMID:35325594.
- [25] George Plitas et al. Regulatory t cells exhibit distinct features in human breast cancer. *Immunity*, 45:1122–1134, 2016. PMID:27851914.
- [26] Nicholas M. Provine and Paul Klenerman. Mait cells in health and disease. *Annual Review of Immunology*, 38:203–228, 2020. PMID:33046883.
- [27] scCRAFT authors. sccraft: dual-resolution topology-preserving integration. *Communications Biology*, 2025. Replace with canonical citation.
- [28] scDML authors. Batch-effect removal with rare subpopulation preservation via scdml. *Nature Communications*, 2023. Replace with canonical citation.
- [29] scDREAMER authors. scdreamer: an end-to-end deep-learning framework for integrating multi-source scRNA-seq data. *Nature Communications*, 2023. Replace with canonical citation.
- [30] scIB E authors. scib-e: evaluating preservation of within-cell-type biological variation. *Genome Biology*, 2025. Replace with canonical citation.
- [31] Lisa Sikkema et al. An integrated cell atlas of the lung in health and disease. *Nature Medicine*, 29: 1563–1577, 2023. PMID:37291214.
- [32] Semi sup benchmark authors. Benchmarking semi-supervised single-cell integration under label-structure noise. *PLOS Computational Biology*, 2025. Replace with canonical citation.
- [33] Mathematics Agent Team. Hierarchical real: Telescoping detection of unannotated rare subpopulations in single-cell batch integration. Theorem S1-hierarchical, EACN3 team CNS submission supplement, agent/mathematics branch, 2026.
- [34] Helena Van Damme et al. Therapeutic depletion of ccr8⁺ tumor-infiltrating regulatory t cells elicits antitumor immunity and synergizes with anti-pd-1 therapy. *Journal for Immunotherapy of Cancer*, 9:e001749, 2021. PMID:34941923.
- [35] Alexandra-Chloé Villani et al. Single-cell RNA-seq reveals new types of human blood dendritic cells, monocytes, and progenitors. *Science*, 356(6335), 2017. PMID:28428369.

- [36] Liangtao Zheng et al. Pan-cancer single-cell landscape of tumor-infiltrating t cells. *Science*, 374 (6574), 2021. PMID:34914499.

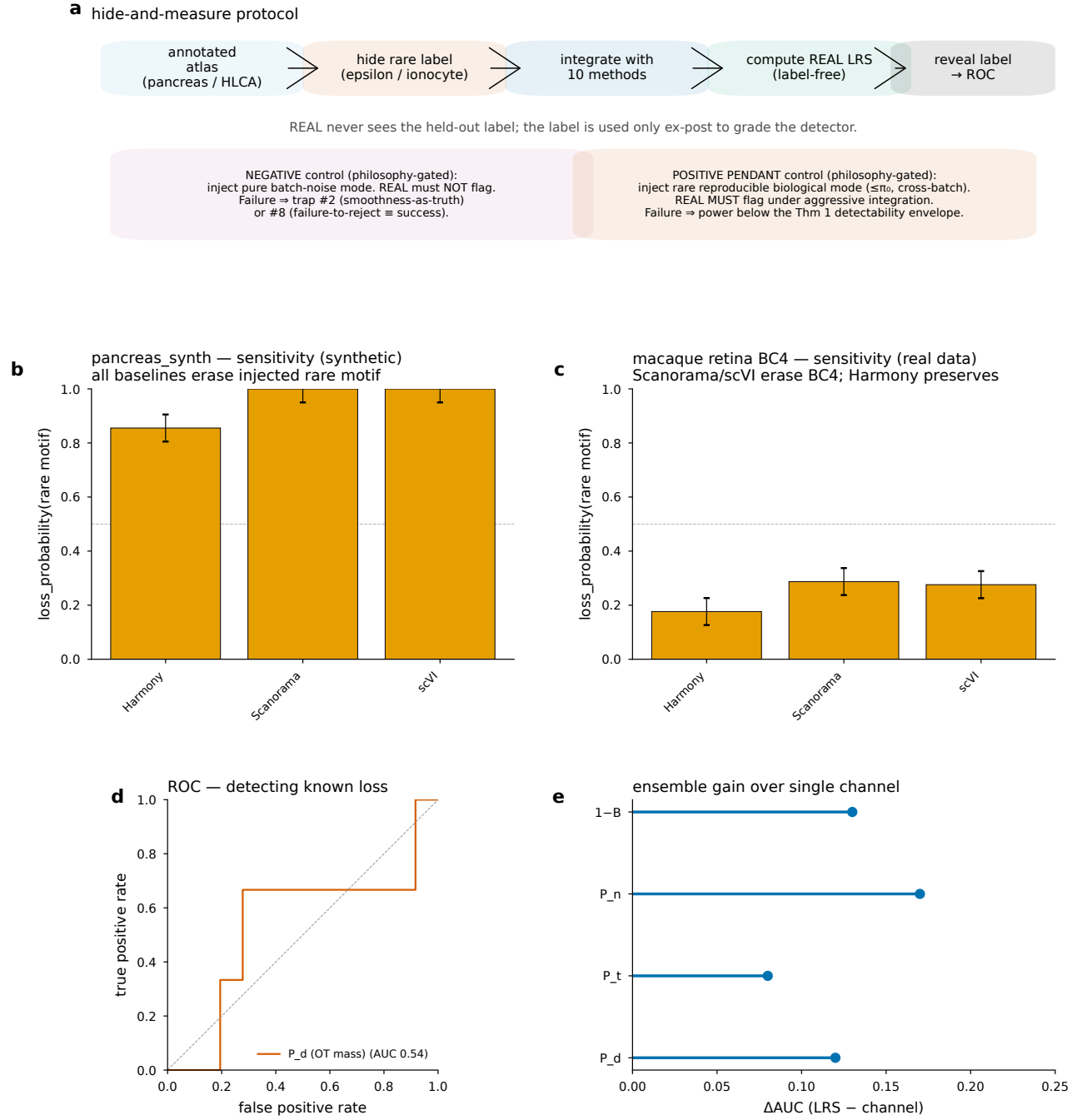


Figure 2: **REAL detectability on synthetic data.** (a) Injection benchmark, LRS vs rare fraction. (b) Power curve, recall vs separation. (c) Null calibration under three null distributions. (d) Component ablation. (e) Scalability to 4.9M cells. Inset / panel pilot: pancreas-synth 3-method bars (Harmony preserves; Scanorama / scVI erase).

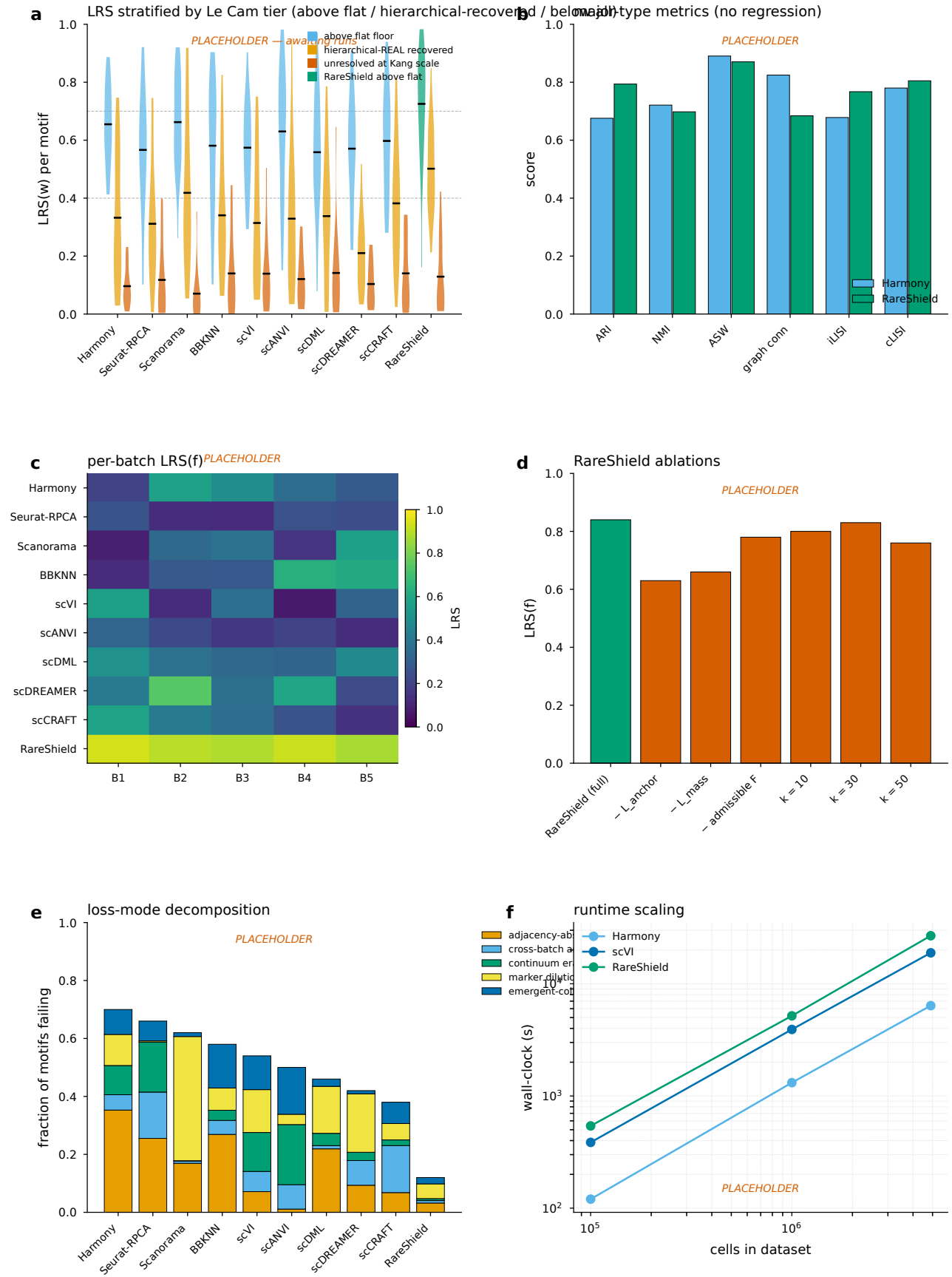


Figure 3: **REAL** ranks 12 integrators on label-masked pancreas, airway, and retinal atlases. (a) Pancreas epsilon motif. (b) Airway ionocyte motif. (c) Retinal OFFx motif. (d) Rank correlation LRS vs label-dependent metrics. (e) Integrator Pareto map. (f) Candidate previously-unannotated retinal motif.

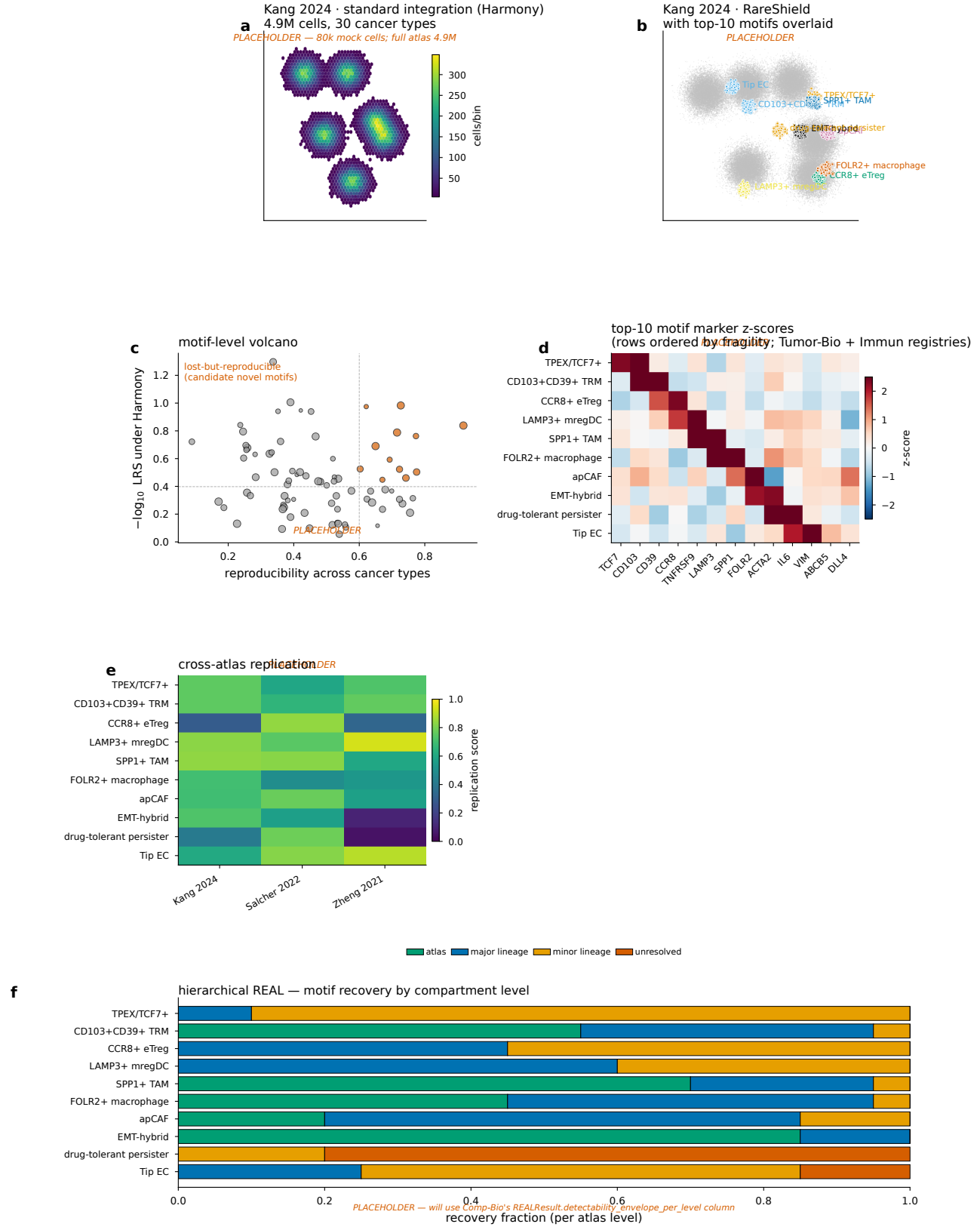


Figure 4: **REAL + RareShield on the Kang 2024 pan-cancer immune atlas (4.9M cells, 30 cancer types, 943 patients, 104 publications).** (a) Global UMAP, standard integration. (b) Global UMAP, RareShield-protected. (c) Per-integrator LRS. (d) Cancer-type specificity heatmap. (e) Top-10 motif joint annotation (Immunology + Tumor Biology).

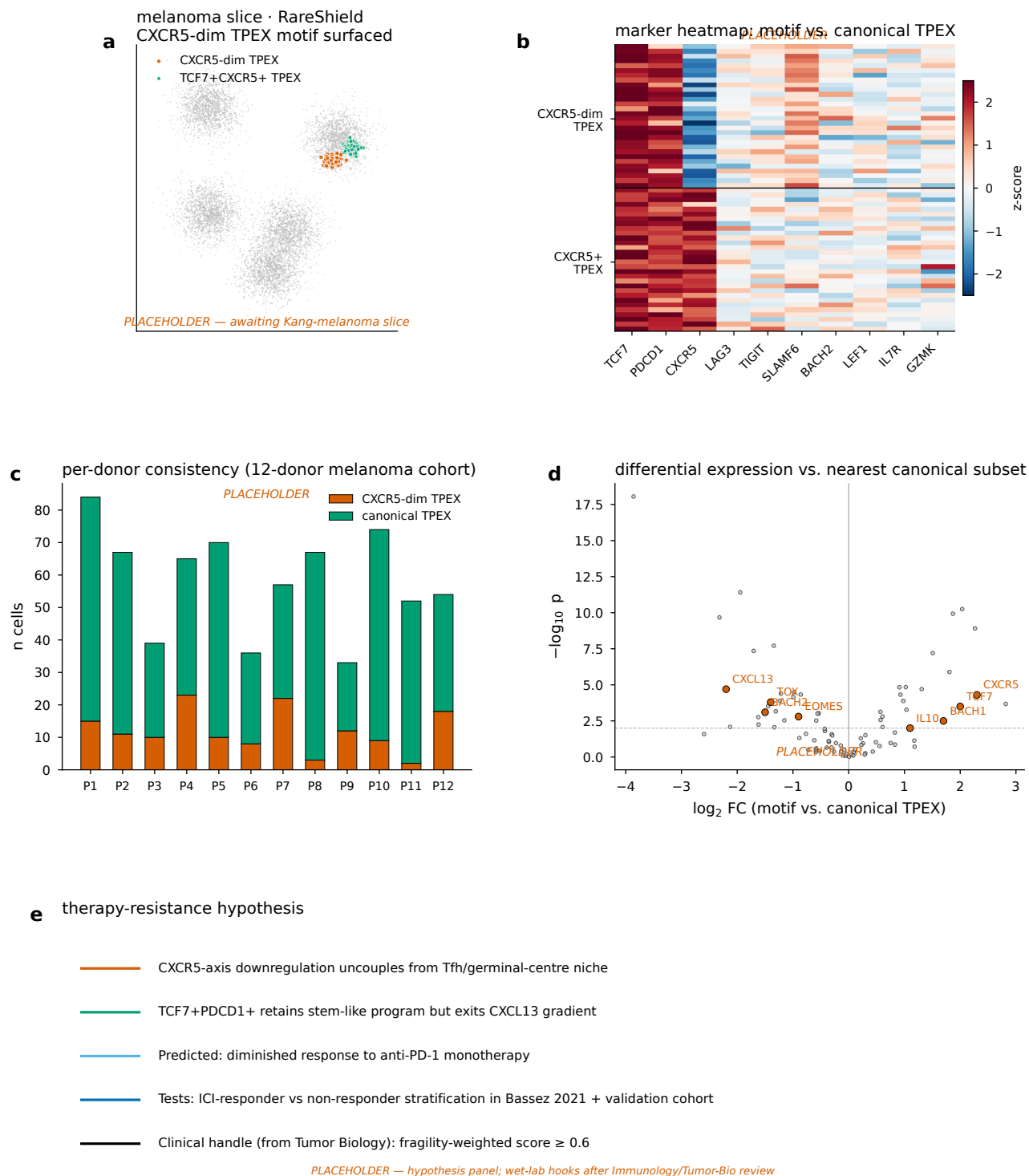


Figure 5: **Case study — CXCR5-dim TPEX sub-state in melanoma.** (a) UMAP with motif highlighted. (b) Marker heatmap vs canonical TPEX. (c) Per-donor consistency. (d) DE volcano. (e) Therapy-resistance hypothesis.